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CHEMICAL BONDING_MEGA_XII



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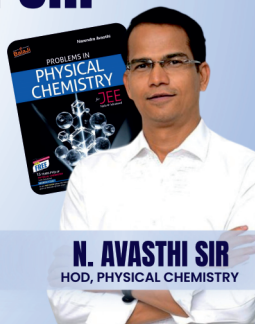
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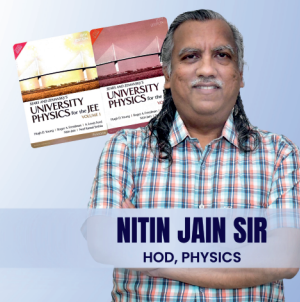


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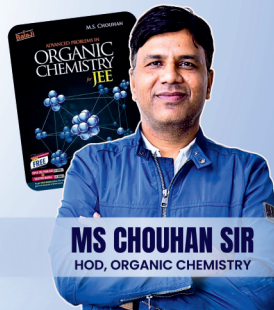


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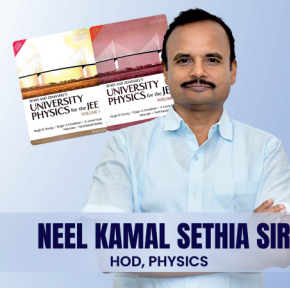


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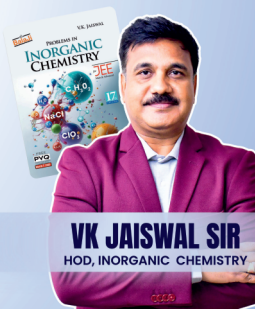


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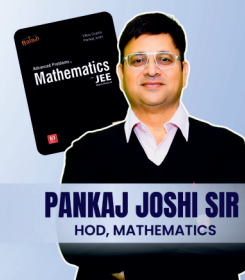


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INORGANIC CHEMISTRY

TARGET IIT JEE MEGA-XII

CHEMICAL BONDING

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EXERCISE-I

Causes of chemical combination :

1. When two atoms combine to form a molecules:

(A) Energy is released
(B) Energy is absorbed

(C) Energy is neither released nor absorbed
(D) Energy may either released or aborbed
2. The combination of atoms occur because they want :

(A) To decrease number of electrons in the outer most orbit
(B)To attain an inert gas configuration

(C) To increase number of electrons in the outer most orbit
(D) To attain 18 electrons in the outermost orbit
3. Which condition favours the bond formation:

(A) Maximum attraction and maximum potential energy
(B) Minimum attraction and minimum potential energy

(C) Minimum potential energy and maximum attraction
(D) None of the above

Ionic bond :

4. An electrovalent bond or ionic bond is formed between:

(A) Two electronegative atoms
(B) Two metals

(C) Electropositive and electronegative atoms
(D) Two electropositive atoms
5. Most favourable conditions for electrovalent bonding are:

(A) Low ionisation potential of one atom and high electron affinity of the other atom
(B) High electron affinity and high ionisation potential of both the atoms

(C) Low electron affinity and low ionisation potential of both the atoms
(D) High ionisation potential of one atom and low electrons afinity of the other atom
6. The electonegativity of Cesium is 0.7 and fluorine is 4.0. The bond formed between the two is:

(A) Covalent
(B) Electrovalent/ionic
(C) Coordinate
(D) Metallic
7. Electrovalent bond or ionic bond is formed by:

(A) Sharing of electons
(B) Donaton of electrons

(C) Transfer of electrons
(D) None of these
8. Element X is strongly electropositive and is strongly electronegative. Both are univalent. The compound formed would be:

(A) $X^+ Y^-$
(B) $X - Y$
(C) $X^- Y^+$
(D) $X \rightarrow Y$
9. As compared to covalent compounds electrovalent compounds generally possess:

(A) High m.p. and high b.p.
(B) Low m.p. and low b.p.

(C) Low m.p. and high b.p.
(D) high m.p. and low b.p.
10. Which has the highest lattice energy?

(A) LiF
(B) LiCl
(C) NaCl
(D) MgO
11. Solid NaCl is a bad conductor of electricity because:

(A) In solid NaCl there are no ions
(B) Solid NaCl is covalent

(C) In solid NaCl there is no mobility of ions
(D) In solid NaCl there are no electrons
12. Electrovalent compounds or ionic compounds do not show isomerism. The reason is:

(A) Presence of ions
(B) Strong electrostatic force of attraction

(C) Brittleness
(D) Non-directional nature of ionic bond

13. Molten sodium chloride conducts electricity due to the presence of :
 (A) Free electrons (B) Free molecules
 (C) Free sodium and chloride atoms (D) Free sodium and chloride ions
14. Choose **incorrect** statement about ionic compounds :
 (A) An ionic solid does not conduct electricity
 (B) They are more soluble in polar solvents like water
 (C) Their enthalpy of formation (ΔH_f) is positive
 (D) They generally have higher melting point than covalent compounds

Covalent Bond :

15. Which of the following statement is not true about covalent compounds:
 (A) They may exhibit space isomerism (B) They have low melting and boiling points
 (C) They show ionic reactions (D) They show molecular reactions
16. In a triple bond there is sharing of:
 (A) 3 electrons (B) 4 electrons (C) Several electrons (D) 6 electrons
17. Which of the following bonds will have directional character:
 (A) Ionic bond (B) Metallic bond (C) Covalent bond (D) None
18. The type of bond formed between two electronegative atoms would be:
 (A) Covalent (B) Ionic (C) Coordinate (D) All the above
19. Elements have electronegativities 2.5 and 4.0, bond formed between them would be:
 (A) Ionic (B) Covalent (C) Co-ordinate (D) Metallic
20. Which of the following is not a characteristic of a covalent compound:
 (A) It has low melting point and boiling point
 (B) Covalent bond is formed between two atoms having very small electronegativity difference
 (C) They have no definite geometry
 (D) They are generally insoluble in water
21. H_3BO_3 is
 (A) monobasic weak Lewis acid (B) monobasic weak Bronsted acid
 (C) monobasic strong Lewis acid (D) tribasic and weak Bronsted acid
22. Which one of the following is smallest in size ?
 (A) N^{3-} (B) O^{2-} (C) F^- (D) Na^+

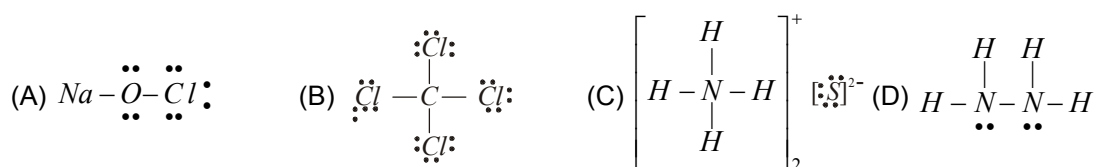
Coordinate Bond :

23. In Co-ordinate bond, the acceptor atoms must essentially contain in its valency shell an orbital:
 (A) With paired electron (B) With single electron
 (C) With no electron (D) With three electron
24. Correct statement regarding this reaction

$$\text{BF}_3 + \text{NH}_3 \longrightarrow [\text{F}_3\text{B} \leftarrow \text{NH}_3]$$

 (A) Hybridisation of N is changed (B) Hybridisation of B is changed
 (C) NH_3 act as a Lewis base (D) (B) & (C) both
25. The correct statement for the reaction : $\text{NH}_3 + \text{H}^+ \rightarrow \text{NH}_4^+$
 (A) Hybridisation state is changed (B) Bond angle increases
 (C) NH_3 act as a Lewis acid (D) Electron geometry is changed

26. The bonds present in N_2O_5 are:
 (A) Only ionic (B) Covalent & coordinate
 (C) Only covalent (D) Covalent & ionic
27. The pair of compounds which can form a co-ordinate bond is:
 (A) $(CH_3)_3B$ and $(CH_3)_3N$ (B) HCl and HBr
 (C) BF_3 and NH_3 (D) (A) and (C) both
28. In which compound coordinate bond is present:
 (A) NH_3 (B) NH_4OH (C) H_2O (D) HCl
29. Which of the following molecules does not have coordinate bonds?
 (A) H_3O^+ (B) NO_3^- (C) O_3 (D) CO_3^{2-}
30. Which of the following Lewis diagrams is(are) incorrect ?

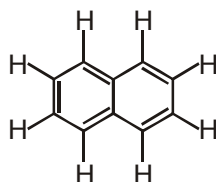


Valence Bond Theory (VBT) :

31. A sigma bond is formed by the overlapping of :
 (A) s-s orbital alone
 (B) s and p orbitals alone
 (C) s-s, s-p or p-p orbitals along internuclear axis
 (D) p-p orbital along the sides
32. Which overlapping is involved in HCl molecule:
 (A) s-s overlap (B) p-p overlap (C) s-d overlap (D) s-p overlap
33. which of the following configuration shows second excitation state of Iodine:
 (A) $\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow}$ (B) $\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow}$
 (C) $\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow}$ (D) $\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow}$
34. Which of the following compound is formed in the second excitation state of sulphur atom:
 (A) SF_4 (B) SF_6 (C) SF_2 (D) None
35. The strength of bonds by $2s - 2s$, $2p - 2p$ and $2p - 2s$ overlapping has the order:
 (A) $s - s > p - p > s - p$ (B) $s - s > p - s > p - p$
 (C) $p - p > s - p > s - s$ (D) $p - p > s - s > p - s$
36. The triple bond in ethyne is made up of:
 (A) Three sigma bonds (B) Three π - bonds
 (C) One sigma and two π bonds (D) Two sigma and one π bond
37. In which of the excitation state of chlorine ClF_3 is formed:
 (A) In ground state (B) In third excitation state
 (C) In first excitation state (D) In second excitation state
38. Variable covalency is exhibited by:
 (A) P and S (B) N and O (C) N and P (D) F and Cl

39. Which is not characteristic of π -bond :
- (A) π -bond is formed when a sigma bond already formed
 (B) π -bond are formed from hybrid orbitals
 (C) π - bond may be formed by the overlapping of p-orbitals
 (D) π - bond results from lateral overlap of atomic orbital
40. π bond is formed :
- (A) By overlapping of hybridised orbital
 (B) Overlapping of axial p-p orbitals
 (C) Head on overlapping of s – s orbitals
 (D) By p – p lateral overlapping
41. Number of σ and π bonds present in $\text{CH}_3 - \text{CH} = \text{CH} - \text{C} \equiv \text{CH}$ are:
- (A) $10\sigma, 3\pi$ (B) $10\sigma, 2\pi$ (C) $9\sigma, 2\pi$ (D) $8\sigma, 3\pi$
42. Which of the following fluorides does not exists?
- (A) NF_5 (B) PF_5 (C) AsF_5 (D) SbF_5
43. p-p overlapping will be observed in the molecules of -
- (A) H_2 (B) HBr (C) HCl (D) Cl_2
44. Nitrogen does not form NF_5 because:
- (A) Nitrogen is member of V group
 (B) It contains no empty d-orbital
 (C) The bond energy of $\text{N} \equiv \text{N}$ is very high
 (D) Inert pair effect exists in the molecule
45. The correct order of bond length is
- (A) $\text{C} - \text{C} < \text{C} = \text{C} < \text{C} \equiv \text{C}$ (B) $\text{C} \equiv \text{C} < \text{C} = \text{C} < \text{C} - \text{C}$
 (C) $\text{C} = \text{C} < \text{C} \equiv \text{C} < \text{C} - \text{C}$ (D) $\text{C} = \text{C} < \text{C} - \text{C} < \text{C} \equiv \text{C}$
46. Which of the following statements is incorrect ?
- (A) Double bonds is shorter than a single bond
 (B) σ – bond is weaker than π bond
 (C) Double bond is stronger than a single bond
 (D) Covalent bond is stronger than a hydrogen bond
47. Fluorine molecule is formed by:
- (A) the axial p-p orbital overlap (B) the sideways p-p orbital overlap
 (C) the s-s orbital overlap (D) the s-p orbital overlap
48. Consider the following statements:
- I. A sigma (σ) bond is formed when two s-orbitals overlap
 II. A pi(π) bond is formed when two p-orbitals overlap axially
 III. A σ - bond is weaker than π -bond
- Which of the above statements is/are correct?
- (A) I and II (B) II and III (C) I alone (D) II alone
49. Number and type of bonds between two carbon atoms in CaC_2 are :
- (A) one sigma (σ) and one pi (π) bond (B) one σ and two π bonds
 (C) one σ and one and a half π bond (D) one σ bond
50. How many σ - and π - bonds are there in salicylic acid (2-hydroxy benzoic acid)?
- (A) $10\sigma, 4\pi$ (B) $16\sigma, 4\pi$ (C) $18\sigma, 2\pi$ (D) $16\sigma, 2\pi$

51. Strongest bond is formed by the head on overlapping of :
 (A) 2s- and 2p- orbitals (B) 2p- and 2p- orbitals
 (C) 2s- and 2s- orbitals (D) All
52. The ratio of σ and π bonds in benzene is :
 (A) 2 (B) 6 (C) 4 (D) 8
53. Number of π -bonds and σ -bonds in the following structure is



- (A) 6, 19 (B) 4, 20 (C) 5, 19 (D) 5, 20

Hybridisation :

54. In the protonation of H_2O , change occurs in :
 (A) Hybridisation state of oxygen (B) Shape of molecule
 (C) Hybridisation and shape both (D) None
55. The d-orbitals involved in sp^3d hybridisation is :
 (A) $\text{d}_{x^2-y^2}$ (B) d_{z^2} (C) d_{xy} (D) d_{xz}
56. A sp^3 hybrid orbital contains:
 (A) $\frac{3}{4}$ s-character (B) $\frac{1}{4}$ p-character (C) $\frac{3}{4}$ p-character (D) $\frac{1}{2}$ s-character
57. In the compound ${}^1\text{CH}_2 = {}^2\text{CH} - {}^3\text{CH}_2 - {}^4\text{CH}_2 - {}^5\text{C} \equiv {}^6\text{CH}$, the $\text{C}_2 - \text{C}_3$ bond formed by the overlapping of :
 (A) $\text{sp} - \text{sp}^2$ (B) $\text{sp}^3 - \text{sp}^3$ (C) $\text{sp} - \text{sp}^3$ (D) $\text{sp}^2 - \text{sp}^3$
58. Which of the set of species have same hybridisation state but different shapes:
 (A) NO_2^+ , NO_2 , NO_2^- (B) ClO_4^- , SF_4 , XeF_4 (C) NH_4^+ , H_3O^+ , OF_2 (D) SO_4^{2-} , PO_4^{3-} , ClO_4^-
59. Which of the following elements can not exhibit sp^3d hybridisation state:
 (1) C (2) P (3) Cl (4) B
 Correct answer is:
 (A) 1, 3 (B) 1, 4 (C) 2, 3 (D) 2, 4
60. In the protonation of NH_3 molecule, following statement is true:
 (A) A covalent bond is formed (B) Hydrogen bond is formed
 (C) Hybridisation state of N is changed (D) Shape of NH_3 molecule is changed
61. The shape of sulphate ion is:
 (A) Hexagonal (B) Square planar (C) Trigonal bipyramidal (D) Tetrahedral
62. In which following set of compound/ion has linear geometry:
 (A) CH_4 , NH_4^+ , BH_4^- (B) CO_3^{2-} , NO_3^- , BF_3 (C) NO_2^+ , CO_2 , N_3^- (D) BeCl_2 , BCl_3 , CH_4

63. Which of the molecule is trigonal bipyramidal :
 (A) BF_3 (B) CH_4 (C) PCl_5 (D) SF_6
64. The type of hybrid orbitals used by chlorine atom in ClO^- , ClO_2^- , ClO_3^- and ClO_4^- is/are:
 (A) sp , sp^2 , sp^3 and sp^3d (B) sp and sp^3
 (C) Only sp^3 (D) Only sp
65. Which of the following having a square planar structure is:
 (A) NH_4^+ (B) BF_4^- (C) XeF_4 (D) CCl_4
66. The shape of I_3^- is :
 (A) Tetrahedral (B) Linear (C) T-shape (D) Trigonal
67. Among the following compounds the one that is polar and has the central atom with sp^3 hybridisation is:
 (A) H_2CO_3 (B) SiF_4 (C) BF_3 (D) HClO_2
68. Which of the following will be octahedral:
 (A) SF_6 (B) BF_4^- (C) PCl_5 (D) XeF_6
69. The shape of methyl cation (CH_3^+) is likely to be:
 (A) linear (B) pyramidal (C) planar (D) spherical
70. The structure of XeF_2 involves hybridization of the type :
 (A) sp^3 (B) sp^3d^3 (C) sp^3d (D) sp^3d^2
71. In the XeF_4 molecule, the Xe atom is in the
 (A) sp^2 -hybridized state (B) sp^3 -hybridised state
 (C) sp^3d -hybridized state (D) sp^3d^2 -hybridized state
72. How many σ - and π - bonds are there in salicylic acid (2-hydroxy benzoic acid)?
 (A) 10σ , 4π (B) 16σ , 4π (C) 18σ , 2π (D) 16σ , 2π
73. Which of the following pairs is (are) isostructural?
 (A) SF_4 and SiF_4 (B) SF_6 and SiF_6^{2-} (C) SiF_6^{2-} and SeF_6^{2-} (D) XeO_6^{4-} and TeF_6^{2-}
74. The structure of XeF_6 is :
 (A) pentagonal bipyramidal (B) trigonal bipyramidal
 (C) capped octahedral (D) square bipyramidal
75. Isostructural species are those which have the same shape and hybridisation. Among the given species identify the isostructural pairs.
 (A) $[\text{NF}_3]$ and $[\text{BF}_3]$ (B) $[\text{BF}_4^-]$ and $[\text{NH}_4^+]$ (C) $[\text{BCl}_3]$ and $[\text{BrCl}_3]$ (D) $[\text{NH}_3]$ and $[\text{NO}_3^-]$
76. The types of hybrid orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ respectively are expected to be
 (A) sp , sp^3 and sp^2 (B) sp , sp^2 and sp^3 (C) sp^2 , sp and sp^3 (D) sp^2 , sp^3 and sp
77. Which of the following angle corresponds to sp^2 hybridisation?
 (A) 90° (B) 120° (C) 180° (D) 109°
78. Which of the following species has tetrahedral geometry?
 (A) BH_4^- (B) NH_2^- (C) CO_3^{2-} (D) H_3O^+
79. Carbon atoms in $\text{C}_2(\text{CN})_4$ are :
 (A) sp -hybridized (B) sp^2 -hybridized
 (C) sp - and sp^2 hybridized (D) sp , sp^2 and sp^3 - hybridized

80. The shapes of IF_5 and IF_7 are respectively :
 (A) square pyramidal and pentagonal bipyramidal
 (B) octahedral and pyramidal
 (C) trigonal bipyramidal and square antiprismatic
 (D) distorted square planar and distorted octahedral
81. CO_2 has the same geometry as :
 (I) HgCl_2 (II) NO_2^- (III) SnCl_4 (IV) C_2H_2
 (A) I and III (B) II and IV (C) I and IV (D) III and IV
82. Which of the following has a geometry different from the other three species (having the same geometry)?
 (A) BF_4^- (B) SO_4^{2-} (C) XeF_4 (D) PH_4^+

Dipole Moment :

83. BeF_2 has zero dipole moment whereas H_2O has dipole moment because:
 (A) Water is linear
 (B) H_2O is bent
 (C) F is more electronegative than O
 (D) Hydrogen bonding is present in H_2O
84. Which of the following molecule have zero dipole moment:
 (A) BF_3 (B) CH_2Cl_2 (C) NF_3 (D) SO_2
85. The dipole moment of NH_3 is:
 (A) Less than dipole moment of NCl_3 (B) Higher than dipole moment of NCl_3
 (C) Equal to the dipole moment of NCl_3 (D) None of these
86. Which of the following species are polar:
 (A) C_6H_6 (B) XeF_2 (C) SO_2 (D) SF_4 (5) SF_6
 Correct answer is:
 (A) (B) & (D) (B) (A), (B) & (5) (C) (A) & (5) (D) (C) & (D)
87. Which set of molecules is polar:
 (A) XeF_4 , IF_7 , SO_3 (B) PCl_5 , C_6H_6 , SF_6
 (C) SnCl_2 , SO_2 , NO_2 (D) CO_2 , CS_2 , C_2H_6
88. Dipole moment is shown by:
 (A) 1, 4 - dichlorobenzene (B) Cis 1, 2 - dichloro ethene
 (C) Trans -1, 2 - dichloro ethene (D) benzene
89. The correct order of dipole moment is:
 (A) $\text{CH}_4 < \text{NF}_3 < \text{NH}_3 < \text{H}_2\text{O}$ (B) $\text{NF}_3 < \text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$
 (C) $\text{NH}_3 < \text{NF}_3 < \text{CH}_4 < \text{H}_2\text{O}$ (D) $\text{H}_2\text{O} < \text{NH}_3 < \text{NF}_3 < \text{CH}_4$
90. Which contains both polar and non-polar bonds?
 (A) NH_4Cl (B) HCN (C) H_2O_2 (D) CH_4

Molecular Orbital Theory (MOT) :

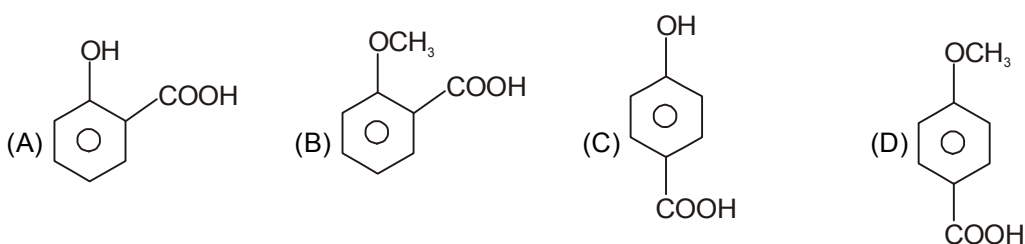
91. The number of antibonding electron pairs in O_2^{2-} molecular ion on the basis of molecular orbital theory is (at no. O = 8):
 (A) 2 (B) 3 (C) 4 (D) 5
92. Which of the following option w.r.t. increasing bond order is correct?
 (A) $\text{NO} < \text{C}_2 < \text{O}_2^- < \text{He}_2^+$ (B) $\text{C}_2 < \text{NO} < \text{He}_2^+ < \text{O}_2^-$
 (C) $\text{He}_2^+ < \text{O}_2^- < \text{NO} < \text{C}_2$ (D) $\text{He}_2^+ < \text{O}_2^- < \text{C}_2 < \text{NO}$

93. The bond-orders of the given species are such that -
 (A) $O_2^- > O_2^+ > O_2^{2-} > O_2$ (B) $O_2^+ > O_2 > O_2^- > O_2^{2-}$
 (C) $O_2^+ > O_2^{2-} > O_2^- > O_2$ (D) $O_2^{2-} > O_2 > O_2^+ > O_2^-$
94. The ion that is isoelectronic with CO and having same bond order is:
 (A) CN^- (B) O_2^+ (C) O_2^- (D) N_2^+
95. Which of the following is paramagnetic:
 (A) O_2^- (B) CN^- (C) CO (D) NO^+
96. In the following which of the two are paramagnetic:
 (A) N_2 (B) CO (C) B_2 (D) NO_2
 Correct answer is:
 (A) (A) & (C) (B) (B) & (C) (C) (C) & (D) (D) (B) & (D)
97. Increasing order of bond length in NO, NO^+ and NO^- is:
 (A) $NO > NO^- > NO^+$ (B) $NO^+ < NO < NO^-$
 (C) $NO < NO^+ < NO^-$ (D) $NO < NO^+ = NO^-$
98. The paramagnetic property of oxygen is well explained by:
 (A) Molecular orbital theory (B) Resonance theory
 (C) Valence bond theory (D) VSEPR theory
99. Which of the following has fractional bond order?
 (A) O_2^{2+} (B) O_2^- (C) F_2^{2-} (D) H_2^-
100. Higher is the bond order, greater is:
 (A) Bond dissociation energy (B) Covalent character
 (C) Bond length (D) Paramagnetism
101. The bond order of CO molecule on the basis of molecular orbital theory is:
 (A) Zero (B) 2 (C) 3 (D) 1
102. Which of the following group of molecules have $2\frac{1}{2}$ bond order:
 (A) N_2^{2-} , O_2^{2-} , CO (B) N_2^+ , O_2^+ , NO (C) C_2^{2-} , BN, O_2 (D) CN^- , NO^+ , O_2^{+2}
103. Which of the following species will have the minimum bond energy:
 (A) N_2 (B) N_2^- (C) N_2^+ (D) N_2^{2-}
104. Which of the following ion has not bond order of 2.5 ?
 (A) O_2^- (B) O_2^+ (C) N_2^+ (D) N_2^-
105. N_2 and O_2 are converted into monoanions, N_2^- and O_2^- respectively. Which of the following statements is wrong ?
 (A) In N_2^- , N-N bond weakens (B) In O_2^- , O-O bond order increases
 (C) In O_2^- , O-O bond order decreases (D) N_2^- becomes paramagnetic

106. Which of the following has maximum bond strength -
 (A) O_2 (B) O_2^+ (C) O_2^- (D) O_2^{2-}

Hydrogen bonding :

107. H_2O boils at higher temperature than H_2S , because it is capable of forming:
 (A) Ionic bonds (B) Covalent bonds
 (C) Hydrogen bonds (D) Metallic bonds
108. Intramolecular H-bond:
 (A) Decreases Volatility (B) Increases melting point
 (C) Increases viscosity (D) Increases vapour pressure
109. Weakest hydrogen bond is:
 (A) $O - H \cdots N$ (B) $S - H \cdots S$ (C) $F - H \cdots F$ (D) $N - H \cdots N$
110. Which of the following has strongest intra molecular hydrogen bonding:



111. The high boiling point of water is due to the presence of:
 (A) Dative bond (B) Covalent bond
 (C) Hydrogen bond (D) Vander waals bond
112. Maximum no. of hydrogen bonds formed by a water molecule in ice is:
 (A) 4 (B) 3 (C) 2 (D) 1
113. KF combines with HF to form KHF_2 . The compound contains the species:
 (A) K^+ , F^- and H^+ (B) K^+ , F^- and HF
 (C) K^+ and $[HF_2]^-$ (D) $[KHF]^+$ and F_2
114. The boiling point of a compound is raised by:
 (A) intermolecular hydrogen bonding (B) high volatility
 (C) intramolecular hydrogen bonding (D) non-polarity
115. The boiling point of p-nitrophenol is higher than that of o-nitrophenol because:
 (A) NO_2 group at p-position behaves in a different way from that at o-position
 (B) intramolecular hydrogen bonding exists in p-nitrophenol
 (C) there is intermolecular hydrogen bonding in p-nitrophenol
 (D) p-nitrophenol has a higher molecular weight than o-nitrophenol
116. Which of the following compounds show intramolecular hydrogen bonding:
 (A) o-nitrophenol (B) p-nitrophenol (C) phenol (D) salicylaldehyde
 (A) 1 & 2 (B) 1 & 3 (C) 1 & 4 (D) 2 & 3

Integer types questions :

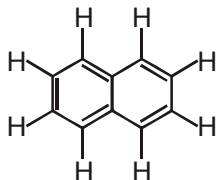
117. What is covalency of chlorine atom in second excited state ?
118. Sum of σ and π bonds in NH_4^+ cation is

119. The molecule ML_x is planar with 6 electron pairs around M in the valence shell. The value of x is :
120. Consider the following species NO_3^- , SO_4^{2-} , ClO_3^- , SO_3 , PO_4^{3-} , XeO_3 , CO_3^{2-} , SO_3^{2-} .
Then calculate value of $|x - y|$, where
x : Total number of species which have bond order 1.5 or greater than 1.5
y : Total number of species which have bond order less than 1.5
121. In IF_6^- and TeF_5^- , sum of axial d-orbitals which are used in hybridisation in both species.
122. Among the following total no. of planar species is :
(i) SF_4 (ii) BrF_3 (iii) XeF_2 (iv) IF_5 (v) SbF_4^- (vi) SF_5^-
(vii) SeF_3^+ (viii) CH_3^+ (ix) PCl_4^+
123. Calculate the value of $X - Y$, for $XeOF_4$. (X = Number of σ bond pair and Y = Number of lone pair on central atom)
124. Find out number of transformation among following which involves the change of hybridisation of underlined atom.
(a) $H_2\underline{O} + H^+ \longrightarrow H_3\underline{O}^+$ (b) $NH_3 + \underline{B}F_3 \longrightarrow NH_3 \cdot \underline{B}F_3$
(c) $\underline{Xe}F_6 \longrightarrow \underline{Xe}F_5^+ + F^-$ (d) $2\underline{P}Cl_5 \longrightarrow \underline{P}Cl_4^+ + \underline{P}Cl_6^-$
(e) $\underline{C}H_3 - CH_3 \longrightarrow \underline{C}H_3^+ + CH_3^-$
125. Consider the following six changes
(i) $NO \longrightarrow NO^+$ (ii) $O_2^- \longrightarrow O_2^{2-}$ (iii) $O_2 \longrightarrow O_2^+$ (iv) $NO^+ \longrightarrow NO^-$
(v) $NO^+ \longrightarrow NO^{2+}$ (vi) $CO \longrightarrow CO^+$
Then calculate value of " $c^3 - b^2 - a$ ", where a, b and c are total number of transformations in which magnetic property will be changed, bond order increases and bond order decreases respectively.
126. Total number of angle in $SeCl_4$ which are less than 90° .

EXERCISE-II

NCERT-EXAMPLER PROBLEMS

Single Choice Questions :

- Isostructural species are those which have the same shape/geometry. Among the given species identify the isostructural pairs.
(A) $[\text{NF}_3 \text{ and } \text{BF}_3]$ (B) $[\text{BF}_4^- \text{ and } \text{NH}_4^+]$ (C) $[\text{BCl}_3 \text{ and } \text{BrCl}_3]$ (D) $[\text{NH}_3 \text{ and } \text{NO}_3^-]$
 - Polarity in a molecule and hence the dipole moment depends primarily on electronegativity of the constituent atoms and shape of a molecule. Which of the following has the highest dipole moment?
(A) CO_2 (B) HI (C) H_2O (D) SO_2
 - The types of hybrid orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ respectively are expected to be
(A) sp , sp^3 and sp^2 (B) sp , sp^2 and sp^3 (C) sp^2 , sp and sp^3 (D) sp^2 , sp^3 and sp
 - Hydrogen bonds are formed in many compounds e.g., H_2O , HF , NH_3 . The boiling point of such compounds depends to a large extent on the strength of hydrogen bond and the number of hydrogen bonds. The correct decreasing order of the boiling points of above compounds is
(A) $\text{HF} > \text{H}_2\text{O} > \text{NH}_3$ (B) $\text{H}_2\text{O} > \text{HF} > \text{NH}_3$ (C) $\text{NH}_3 > \text{HF} > \text{H}_2\text{O}$ (D) $\text{NH}_3 > \text{H}_2\text{O} > \text{HF}$
 - In PO_4^{3-} ion the formal charge on the oxygen atom of P–O bond is
(A) + 1 (B) – 1 (C) – 0.75 (D) + 0.75
 - In NO_3^- ion, the number of bond pairs and lone pairs of electrons on nitrogen atom are
(A) 2, 2 (B) 3, 1 (C) 1, 3 (D) 4, 0
 - Which of the following species has tetrahedral geometry?
(A) BH_4^- (B) NH_2^- (C) CO_3^{2-} (D) H_3O^+
 - Number of π -bonds and σ -bonds in the following structure is
(A) 6, 19
(B) 4, 20
(C) 5, 19
(D) 5, 20
- 
- Which molecule/ion out of the following does not contain unpaired electrons?
(A) N_2^+ (B) O_2 (C) O_2^{2-} (D) B_2
 - In which of the following molecule/ion all the bonds are not equal?
(A) XeF_4 (B) BF_4^- (C) C_2H_4 (D) SiF_4
 - In which of the following substances will hydrogen bond be strongest?
(A) HCl (B) H_2O (C) HI (D) H_2S
 - If the electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$, the four electrons involved in chemical bond formation will be _____
(A) $3p^6$ (B) $3p^6, 4s^2$ (C) $3p^6, 3d^2$ (D) $3d^2, 4s^2$
 - Which of the following angle corresponds to sp^2 hybridisation?
(A) 90° (B) 120° (C) 180° (D) 109°

The electronic configuration of three elements, A, B and C are given below. Answer the questions 14 to 17 on the basis of these configurations.

A	1s ²	2s ²	2p ⁶		
B	1s ²	2s ²	2p ⁶	3s ²	3p ³
C	1s ²	2s ²	2p ⁶	3s ²	3p ⁵

14. Stable form of A may be represented by the formula :
 (A) A (B) A₂ (C) A₃ (D) A₄
15. Stable form of C may be represented by the formula
 (A) C (B) C₂ (C) C₃ (D) C₄
16. The molecular formula of the compound formed from B and C will be
 (A) BC (B) B₂C (C) BC₂ (D) BC₃
17. The bond between B and C will be
 (A) Ionic (B) Covalent (C) Hydrogen (D) Coordinate
18. Which of the following order of energies of molecular orbitals of N₂ is correct?
 (A) $(\pi 2p_y) < (\sigma 2p_z) < (\pi^* 2p_x) = (\pi^* 2p_y)$ (B) $(\pi 2p_y) > (\sigma 2p_z) > (\pi^* 2p_x) = (\pi^* 2p_y)$
 (C) $(\pi 2p_y) < (\sigma 2p_z) > (\pi^* 2p_x) = (\pi^* 2p_y)$ (D) $(\pi 2p_y) > (\sigma 2p_z) < (\pi^* 2p_x) = (\pi^* 2p_y)$
19. Which of the following statement is not correct from the view point of molecular orbital theory?
 (A) Be₂ is not a stable molecule.
 (B) He₂ is not stable but He₂⁺ is expected to exist.
 (C) Bond strength of N₂ is maximum amongst the homonuclear diatomic molecules belonging to the second period.
 (D) The order of energies of molecular orbitals in N₂ molecules is
 $\sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$
20. Which of the following options represents the correct bond order :
 (A) O₂⁻ > O₂ > O₂⁺ (B) O₂⁻ < O₂ < O₂⁺ (C) O₂⁻ > O₂ < O₂⁺ (D) O₂⁻ < O₂ > O₂⁺
21. The electronic configuration of the outer most shell of the most electronegative element is
 (A) 2s²2p⁵ (B) 3s²3p⁵ (C) 4s²4p⁵ (D) 5s²5p⁵
22. Amongst the following elements whose electronic configurations are given below, the one having the highest ionisation enthalpy is
 (A) [Ne]3s²3p¹ (B) [Ne]3s²3p³ (C) [Ne]3s²3p² (D) [Ar]3d¹⁰4s²4p³

Assertion and Reason Type :

23. **Assertion (A) :** Sodium chloride formed by the action of chlorine gas on sodium metal is a stable compound.
Reason (R) : This is because sodium and chloride ions acquire octet in sodium chloride formation.
 (A) A and R both are correct, and R is the correct explanation of A.
 (B) A and R both are correct, but R is not the correct explanation of A.
 (C) A is true but R is false.
 (D) A and R both are false.

24. **Assertion (A) :** Though the central atom of both NH_3 and H_2O molecules are sp^3 hybridised, yet $\text{H}-\text{N}-\text{H}$ bond angle is greater than that of $\text{H}-\text{O}-\text{H}$.
Reason (R) : This is because nitrogen atom has one lone pair and oxygen atom has two lone pairs.
 (A) A and R both are correct, and R is the correct explanation of A.
 (B) A and R both are correct, but R is not the correct explanation of A.
 (C) A is true but R is false.
 (D) A and R both are false.
25. **Assertion (A) :** Among the two $\text{O}-\text{H}$ bonds in H_2O molecule, the energy required to break the first $\text{O}-\text{H}$ bond and the other $\text{O}-\text{H}$ bond is the same.
Reason (R) : This is because the electronic environment around oxygen is the same even after breakage of one $\text{O}-\text{H}$ bond.
 (A) A and R both are correct, and R is the correct explanation of A.
 (B) A and R both are correct, but R is not the correct explanation of A.
 (C) A is true but R is false.
 (D) A and R both are false.

One or more than one correct :

26. Which of the following have identical bond order?
 (A) CN^- (B) NO^+ (C) O_2^- (D) O_2^{2-}
27. Which of the following attain the linear structure
 (A) BeCl_2 (B) NCO^- (C) NO_2 (D) CS_2
28. CO is isoelectronic with
 (A) NO^+ (B) N_2 (C) SnCl_2 (D) NO_2^-
29. Which of the following species have the same shape ?
 (A) CO_2 (B) CCl_4 (C) O_3 (D) NO_2^-
30. Which of the following statements are correct about CO_3^{2-} ?
 (A) The hybridisation of central atom is sp^2 .
 (B) Its resonance structure has one $\text{C}-\text{O}$ single bond and two $\text{C}=\text{O}$ double bonds.
 (C) The average formal charge on each oxygen atom is 0.67 units.
 (D) All $\text{C}-\text{O}$ bond lengths are equal.
31. Diamagnetic species are those which contain no unpaired electrons. Which among the following are diamagnetic?
 (A) N_2 (B) N_2^{2-} (C) O_2 (D) O_2^{2-}
32. Species having same bond order are :
 (A) N_2 (B) N_2^- (C) F_2^+ (D) O_2^-
33. Which of the following statements are not correct?
 (A) NaCl being an ionic compound is a good conductor of electricity in the solid state.
 (B) In canonical structures there is a difference in the arrangement of atoms.
 (C) Hybrid orbitals form stronger bonds than pure orbitals.
 (D) VSEPR Theory can explain the square planar geometry of XeF_4 .

Comprehension (Q. No. 34 to 37) :

Comprehension given below is followed by some multiple choice questions. Each question has one correct option. Choose the correct option.

Molecular orbitals are formed by the overlap of atomic orbitals. Two atomic orbitals combine to form two molecular orbitals called bonding molecular orbitals (BMO) and anti bonding molecular orbital (ABMO). Energy of anti bonding orbital is raised above the parent atomic orbitals that have combined and the energy of the bonding orbital is lowered than the parent atomic orbitals. Energies of various molecular orbitals for elements hydrogen to nitrogen increase in the order :

$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$ and for oxygen and fluorine order of energy of molecular orbitals is given below :

$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

Different atomic orbitals of one atom combine with those atomic orbitals of the second atom which have comparable energies and proper orientation. Further, if the overlapping is head on, the molecular orbital is called 'Sigma', (σ) and if the overlap is lateral, the molecular orbital is called 'pi', (π). The molecular orbitals are filled with electrons according to the same rules as followed for filling of atomic orbitals. However, the order for filling is not the same for all molecules or their ions. Bond order is one of the most important parameters to compare the strength of bonds.

34. Which of the following statements is correct?
 (A) In the formation of dioxygen from oxygen atoms 10 molecular orbitals will be formed.
 (B) All the molecular orbitals in the dioxygen will be completely filled.
 (C) Total number of bonding molecular orbitals will not be same as total number of anti bonding orbitals in dioxygen.
 (D) Number of filled bonding orbitals will be same as number of filled anti bonding orbitals.
35. Which of the following molecular orbitals has maximum number of nodal planes?
 (A) $\sigma^* 1s$ (B) $\sigma^* 2p_z$ (C) $\pi 2p_x$ (D) $\pi^* 2p_y$
36. Which of the following pair is expected to have the same bond order?
 (A) O_2, N_2 (B) O_2^+, N_2^- (C) O_2^-, N_2^+ (D) O_2^-, N_2^-
37. In which of the following molecules, $\sigma 2p_z$ molecular orbital is filled after $\pi 2p_x$ and $\pi 2p_y$ molecular orbitals?
 (A) O_2 (B) Ne_2 (C) N_2 (D) F_2

MATCH THE COLUMN

38. Match the species in Column I with the type of hybrid orbitals in Column II.

Column - I

- (A) SF_4
 (B) IF_5
 (C) NO_2^+
 (D) NH_4^+

Column - II

- (P) sp^3d^2
 (Q) d^2sp^3
 (R) sp^3d
 (S) sp^3
 (T) sp

39. Match the species in Column I with the geometry / shape in Column II

Column - I

- (A) H_3O^+
 (B) $HC \equiv CH$
 (C) ClO_2^-
 (D) NH_4^+

Column - II

- (P) Linear
 (Q) Angular
 (R) Tetrahedral
 (S) Trigonal bipyramidal
 (T) Pyramidal

40. Match the species in Column I with the bond order in Column II.

Column - I

- (A) NO
- (B) CO
- (C) O_2^-
- (D) O_2

Column - II

- (P) 1.5
- (Q) 2.0
- (R) 2.5
- (S) 3.0

41. Match the items given in Column I with examples given in Column II.

Column - I

- (A) Hydrogen bond
- (B) Resonance
- (C) Ionic solid
- (D) Covalent solid

Column - II

- (P) C
- (Q) LiF
- (R) H_2
- (S) HF
- (T) O_3

42. Match the shape of molecules in Column I with the type of hybridisation in Column II

Column - I

- (A) Tetrahedral
- (B) Trigonal
- (C) Linear

Column - II

- (P) sp^2
- (Q) sp
- (R) sp^3

EXERCISE - III

(Question asked in previous JEE-MAIN)

1. The C—C bond length is maximum in : [JEE-Main-Online 2020]
 (A) diamond (B) C_{70} (C) C_{60} (D) graphite
2. The oxide **without** nitrogen-nitrogen bond is : [JEE-Main-Online 2020]
 (A) N_2O_3 (B) N_2O_5 (C) N_2O (D) N_2O_4
3. The number of sp^2 hybrid orbitals in a molecule of benzene is [JEE-Main-Online 2020]
 (A) 6 (B) 24 (C) 18 (D) 12
4. If the magnetic moment of a dioxygen species is 1.73 B.M, it may be : [JEE-Main-Online 2020]
 (A) O_2 , O_2^- or O_2^+ (B) O_2 or O_2^- (C) O_2 or O_2^+ (D) O_2^- or O_2^+
5. Arrange the following bonds according to their average bond energies in descending order : [JEE-Main-Online 2020]
 $C - Cl$, $C - Br$, $C - F$, $C - I$
 (A) $C - F > C - Cl > C - Br > C - I$ (B) $C - I > C - Br > C - Cl > C - F$
 (C) $C - Br > C - I > C - Cl > C - F$ (D) $C - Cl > C - Br > C - I > C - F$
6. The predominant intermolecular forces present in ethyl acetate, a liquid, are : [JEE-Main-Online 2020]
 (A) London dispersion and dipole-dipole.
 (B) London dispersion, dipole-dipole and hydrogen bonding.
 (C) Dipole-dipole and hydrogen bonding.
 (D) Hydrogen bonding and London dispersion.
7. The relative strength of interionic/intermolecular forces in decreasing order is : [JEE-Main-Online 2020]
 (A) dipole-dipole > ion-dipole > ion-ion (B) ion-dipole > ion-ion > dipole-dipole
 (C) ion-dipole > dipole-dipole > ion-ion (D) ion-ion > ion-dipole > dipole-dipole
8. The dipole moments of CCl_4 , $CHCl_3$ and CH_4 are in the order : [JEE-Main-Online 2020]
 (A) $CHCl_3 < CH_4 = CCl_4$ (B) $CH_4 = CCl_4 < CHCl_3$
 (C) $CCl_4 < CH_4 < CHCl_3$ (D) $CH_4 < CCl_4 < CHCl_3$
9. The bond order and the magnetic characteristics of CN^- are : [JEE-Main-Online 2020]
 (A) $2\frac{1}{2}$, paramagnetic (B) $2\frac{1}{2}$, diamagnetic (C) 3, paramagnetic (D) 3, diamagnetic
10. Match the type of interaction in column A with the distance dependence of their interaction energy in column B : [JEE-Main-Online 2020]

A	B
(I) ion-ion	(a) $\frac{1}{r}$
(II) dipole-dipole	(b) $\frac{1}{r^2}$
(III) London dispersion	(c) $\frac{1}{r^3}$
	(d) $\frac{1}{r^6}$
(A) (I)-(b); (II)-(d); (III)-(c)	(B) (I)-(a); (II)-(b); (III)-(d)
(C) (I)-(a); (II)-(b); (III)-(c)	(D) (I)-(a); (II)-(c); (III)-(d)

11. Among the sulphates of alkaline earth metals, the solubilities of BeSO_4 and MgSO_4 in water, respectively, are : **[JEE-Main-Online 2020]**
 (A) poor and poor (B) High and poor (C) High and high (D) Poor and high
12. The compound that has the largest H–M–H bond angle (M = N, O, S, C), is : **[JEE-Main-Online 2020]**
 (A) H_2O (B) NH_3 (C) H_2S (D) CH_4
13. The shape/structure of $[\text{XeF}_5]^-$ and XeO_3F_2 , respectively, are : **[JEE-Main-Online 2020]**
 (A) pentagonal planar and trigonal bipyramidal (B) octahedral and square pyramidal
 (C) trigonal bipyramidal and pentagonal planar (D) trigonal bipyramidal and trigonal bipyramidal
14. The molecular geometry of SF_6 is octahedral. What is the geometry of SF_4 (including lone pair(s) of electrons, if any) ? **[JEE-Main-Online 2020]**
 (A) Tetrahedral (B) Trigonal bipyramidal
 (C) Pyramidal (D) Square planar
15. In a molecule of pyrophosphoric acid, the number of P–OH, P = O and P–O–P bonds/moiety(ies) respectively are : **[JEE-Main-Online 2020]**
 (A) 2, 4 and 1 (B) 3, 3 and 3 (C) 4, 2 and 0 (D) 4, 2 and 1
16. Of the species, NO , NO^+ , NO^{2+} and NO^- , the one with minimum bond strength is : **[JEE-Main-Online 2020]**
 (A) NO^+ (B) NO (C) NO^{2+} (D) NO^-
17. The structure of PCl_5 in the solid state is : **[JEE-Main-Online 2020]**
 (A) Tetrahedral $[\text{PCl}_4]^+$ and octahedral $[\text{PCl}_6]^-$ (B) Square planar $[\text{PCl}_4]^+$ and octahedral $[\text{PCl}_6]^-$
 (C) Square pyramidal (D) Trigonal bipyramidal
18. The reaction in which the hybridisation of the underlined atom is affected is : **[JEE-Main-Online 2020]**
 (A) $\text{H}_3\underline{\text{P}}\text{O}_2 \xrightarrow{\text{Disproportionation}}$ (B) $\text{H}_2\underline{\text{S}}\text{O}_4 + \text{NaCl} \xrightarrow{420\text{K}}$
 (C) $\underline{\text{N}}\text{H}_3 \xrightarrow{\text{H}^+}$ (D) $\underline{\text{Xe}}\text{F}_4 + \text{SbF}_5 \longrightarrow$
19. The oxidation states of transition metal atoms in $\text{K}_2\text{Cr}_2\text{O}_7$, KMnO_4 and K_2FeO_4 , respectively, are x, y and z. The sum of x, y and z is. **[JEE-Main-Online 2020]**
20. The number of Cl = O bonds in perchloric acid is : **[JEE-Main-Online 2020]**
21. Which among the following species has unequal bond lengths? **[JEE-Main-Online 2021]**
 (A) SF_4 (B) BF_4^- (C) SiF_4 (D) XeF_4
22. The correct order of bond dissociation enthalpy of halogens is **[JEE-Main-Online 2021]**
 (A) $\text{I}_2 > \text{Br}_2 > \text{Cl}_2 > \text{F}_2$ (B) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$
 (C) $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$ (D) $\text{Cl}_2 > \text{F}_2 > \text{Br}_2 > \text{I}_2$
23. According to molecular orbital theory, the species among the following that does not exist is **[JEE-Main-Online 2021]**
 (A) O_2^{2-} (B) He_2^- (C) He_2^+ (D) Be_2

24. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : Dipole-dipole interactions are the only non-covalent interactions, resulting in hydrogen bond formation.
Reason R : Fluorine is the most electronegative element and hydrogen bonds in HF are symmetrical.
 In the light of the above statements, choose the most appropriate answer from the options given below :
[JEE-Main-Online 2021]

- (A) Both A and R are true but R is NOT the correct explanation of A
 (B) Both A and R are true and R is the correct explanation of A
 (C) A is false but R is true
 (D) A is true but R is false

25. Given below are two statements : **[JEE-Main-Online 2021]**

Statement-I : o-Nitrophenol is steam volatile due to intramolecular hydrogen bonding.

Statement-II : o-Nitrophenol has high melting due to hydrogen bonding.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Statement I is false but Statement II is true
 (B) Statement I is true but Statement II is false
 (C) Both Statement I and Statement II is true
 (D) Both Statement I and Statement II is false

26. The correct shape and I – I – I bond angles respectively in I_3^- ion are **[JEE-Main-Online 2021]**

- (A) Distorted trigonal planar ; 135° and 90°
 (B) T-shaped ; 180° and 90°
 (C) Linear ; 180°
 (D) Trigonal planar ; 120°

27. Match List-I with List-II. **[JEE-Main-Online 2021]**

List-I (Molecule)		List-II (Bond order)	
(a)	Ne_2	(i)	1
(b)	N_2	(ii)	2
(c)	F_2	(iii)	0
(d)	O_2	(iv)	3

Choose the correct answer from the options given below :

- (A) (a) $\rightarrow$ (i), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (iv)
 (B) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (iii)
 (C) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (i)
 (D) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (i), (d) $\rightarrow$ (ii)

28. In ${}^1CH_2 = {}^2C = {}^3CH - {}^4CH_3$ molecule, the hybridization of carbon 1, 2, 3 and 4 respectively are :

[JEE-Main-Online 2021]

- (A) sp^2 , sp^3 , sp^2 , sp^3 (B) sp^3 , sp , sp^3 , sp^3
 (C) sp^2 , sp , sp^2 , sp^3 (D) sp^2 , sp^2 , sp^2 , sp^3

29. Which of the following are isostructural pairs? **[JEE-Main-Online 2021]**

- (a) SO_4^{2-} and CrO_4^{2-} (b) $SiCl_4$ and $TiCl_4$ (c) NH_3 and NO_3 (d) BCl_3 and $BrCl_3$
 (A) c and d only (B) a and b only (C) a and c only (D) b and c only

30. Out of the following, which type of interaction is responsible for the stabilisation of α -helix structure of proteins? [JEE-Main-Online 2021]
 (A) vander Waals forces (B) Hydrogen bonding
 (C) Ionic bonding (D) Covalent bonding
31. Amongst the following, the linear species is : [JEE-Main-Online 2021]
 (A) NO_2 (B) Cl_2O (C) O_3
 (D) N_3^-
32. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)** :
Assertion (A) : The H–O–H bond angle in water molecule is 104.5°
Reason (R) : The lone pair-lone pair repulsion of electrons is higher than the bond pair-bond pair repulsion.
 In the light of the above statements, choose the correct answer from the options given below : [JEE-Main-Online 2021]
 (A) **A** is false but **R** is true
 (B) Both **A** and **R** are true but **R** is not the correct explanation of **A**
 (C) **A** is true but **R** is false
 (D) Both **A** and **R** are true and **R** is the correct explanation of **A**
33. Number of paramagnetic oxides among the following given oxides is Li_2O , CaO , Na_2O_2 , KO_2 , MgO and K_2O [JEE-Main-Online 2021]
 (A) 0 (B) 3 (C) 2 (D) 1
34. A central atom in a molecule has two lone pairs of electrons and forms three single bonds. The shape of this molecule is : [JEE-Main-Online 2021]
 (A) see-saw (B) Planar triangular (C) T-shaped (D) trigonal pyramidal
35. Which of the following compound CANNOT act as a Lewis base ? [JEE-Main-Online 2021]
 (A) NF_3 (B) PCl_5 (C) SF_4 (D) ClF_3
36. The oxidation states of nitrogen in NO , NO_2 , and NO_3^- , N_2O are in the order of: [JEE-Main-Online 2021]
 (A) $\text{NO}_3^- > \text{NO}_2 > \text{NO} > \text{N}_2\text{O}$ (B) $\text{NO}_2 > \text{NO}_3^- > \text{NO} > \text{N}_2\text{O}$
 (C) $\text{N}_2\text{O} > \text{NO}_2 > \text{NO} > \text{NO}_3^-$ (D) $\text{NO} > \text{NO}_2 > \text{N}_2\text{O} > \text{NO}_3^-$
37. A hard substance melts at high temperature and is an insulator in both solid and in molten state. This solid is most likely to be a/an : [JEE-Main-Online 2021]
 (A) Ionic solid (B) Molecular solid (C) Metallic solid (D) Covalent solid
38. The correct order of following 3d metal oxides, according to their oxidation numbers is : [JEE-Main-Online 2021]
 (a) CrO_3 (b) Fe_2O_3 (c) MnO_2 (d) V_2O_5
 (e) Cu_2O
 (A) (a) > (c) > (d) > (b) > (e) (B) (d) > (a) > (b) > (c) > (e)
 (C) (a) > (d) > (c) > (b) > (e) (D) (c) > (a) > (d) > (e) > (b)
39. The water having more dissolved O_2 is : [JEE-Main-Online 2021]
 (A) polluted water (B) water at 80°C (C) boiling water (D) water at 4°C
40. The set in which compounds have different nature is : [JEE-Main-Online 2021]
 (A) $\text{Be}(\text{OH})_2$ and $\text{Al}(\text{OH})_3$ (B) $\text{B}(\text{OH})_3$ and $\text{Al}(\text{OH})_3$
 (C) $\text{B}(\text{OH})_3$ and H_3PO_3 (D) NaOH and $\text{Ca}(\text{OH})_2$

41. The hybridisations of the atomic orbitals of nitrogen in NO_2^- , NO_2^+ and NH_4^+ respectively : [JEE-Main-Online 2021]
- (A) sp^3 , sp and sp^2 (B) sp^2 , sp and sp^3 (C) sp^3 , sp^2 and sp (D) sp , sp^2 and sp^3
42. In the following the correct bond order sequence is : [JEE-Main-Online 2021]
- (A) $\text{O}_2 > \text{O}_2^- > \text{O}_2^{2-} > \text{O}_2^+$ (B) $\text{O}_2^{2-} > \text{O}_2^+ > \text{O}_2^- > \text{O}_2$
- (C) $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$ (D) $\text{O}_2^+ > \text{O}_2^- > \text{O}_2^{2-} > \text{O}_2$
43. Identify the species having one π -bond and maximum number of canonical forms from the following : [JEE-Main-Online 2021]
- (A) O_2 (B) SO_3 (C) CO_3^{2-} (D) SO_2
44. Identify the process in which change in the oxidation state is five [JEE-Main-Online 2021]
- (A) $\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{CO}_2$ (B) $\text{Cr}_2\text{O}_7^{2-} \rightarrow 2\text{Cr}^{3+}$ (C) $\text{CrO}_4^{2-} \rightarrow \text{Cr}^{3+}$ (D) $\text{MnO}_4^- + \text{Mn}^{2+}$
45. Match List - I with List - II : [JEE-Main-Online 2021]
- | List - I
(Species) | List - II
(Hybrid Orbitals) |
|-----------------------|--------------------------------|
| (a) SF_4 | (i) sp^3d^2 |
| (b) IF_5 | (ii) d^2sp^3 |
| (c) NO_2^+ | (iii) sp^3d |
| (d) NH_4^+ | (iv) sp^3 |
| | (v) sp |
- Choose the correct answer from the options given below :
- (A) (a)-(iii), (b)-(i), (c)-(v) and (d)-(iv) (B) (a)-(ii), (b)-(i), (c)-(iv) and (d)-(v)
- (C) (a)-(iv), (b)-(iii), (c)-(ii) and (d)-(v) (D) (a)-(i), (b)-(ii), (c)-(v) and (d)-(iii)
46. Which one of the following group-15 hydride is the strongest reducing agent ? [JEE-Main-Online 2021]
- (A) SbH_3 (B) BiH_3 (C) PH_3 (D) AsH_3
47. The oxidation states of 'P' in $\text{H}_4\text{P}_2\text{O}_7$, $\text{H}_4\text{P}_2\text{O}_5$ and $\text{H}_4\text{P}_2\text{O}_6$, respectively, are : [JEE-Main-Online 2021]
- (A) 6, 4 and 5 (B) 5, 3 and 4 (C) 7, 5 and 6 (D) 5, 4 and 3
48. Match List - I with List - II : [JEE-Main-Online 2021]
- | List - I | List - II |
|------------------------------|------------------|
| (a) NaOH | (i) Acidic |
| (b) $\text{Be}(\text{OH})_2$ | (ii) Basic |
| (c) $\text{Ca}(\text{OH})_2$ | (iii) Amphoteric |
| (d) $\text{B}(\text{OH})_3$ | |
| (e) $\text{Al}(\text{OH})_3$ | |
- Choose the most appropriate answer from the option given below :
- (A) (a)-(ii), (b)-(ii), (c)-(iii), (d)-(i), (e)-(iii) (B) (a)-(ii), (b)-(iii), (c)-(ii), (d)-(i), (e)-(iii)
- (C) (a)-(ii), (b)-(ii), (c)-(iii), (d)-(ii), (e)-(iii) (D) (a)-(ii), (b)-(i), (c)-(ii), (d)-(iii), (e)-(iii)
49. Number of $\text{Cl} = \text{O}$ bonds in chlorous acid, chloric acid and perchloric acid respectively are : [JEE-Main-Online 2021]
- (A) 4, 1 and 0 (B) 3, 1 and 1 (C) 1, 1 and 3 (D) 1, 2 and 3
50. The bond order and magnetic behaviour of O_2^- ion are, respectively : [JEE-Main-Online 2021]
- (A) 2 and diamagnetic (B) 1.5 and paramagnetic
- (C) 1 and paramagnetic (D) 1.5 and diamagnetic

51. The interaction energy of London forces between two particles is proportional to r^x , where r is the distance between the particles. The value of x is : [JEE-Main-Online 2021]
 (A) 3 (B) -3 (C) -6 (D) 6
52. Match List - I with List - II : [JEE-Main-Online 2021]
- | List - I
(Species) | List - II
(No. of lone pairs of electrons on the central atom) |
|------------------------------|---|
| (a) XeF_2 | (i) 0 |
| (b) XeO_2F_2 | (ii) 1 |
| (c) XeO_3F_2 | (iii) 2 |
| (d) XeF_4 | (iv) 3 |
- Choose the most appropriate answer from the options given below
- (A) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i) (B) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
 (C) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii) (D) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
53. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Barium carbonate is insoluble in water and is highly stable.
Reason (R) : The thermal stability of the carbonates increases with increasing cationic size.
 Choose the most appropriate answer from the options given below : [JEE-Main-Online 2021]
 (A) Both **(A)** and **(R)** are true and **(R)** is the true explanation of **(A)**.
 (B) Both **(A)** and **(R)** are true but **(R)** is not the true explanation of **(A)**.
 (C) **(A)** is true but **(R)** is false.
 (D) **(A)** is false but **(R)** is true.
54. The number of S = O bonds present in sulphurous acid, peroxodisulphuric acid and pyrosulphuric acid, respectively are : [JEE-Main-Online 2021]
 (A) 2, 3 and 4 (B) 1, 4 and 3 (C) 1, 4 and 4 (D) 2, 4 and 3
55. In polythionic acid, $\text{H}_2\text{S}_x\text{O}_6$ ($x = 3$ to 5) the oxidation state(s) of sulphur is/are : [JEE-Main-Online 2021]
 (A) +5 only (B) 0 and + 5 only (C) + 6 only (D) + 3 and + 5 only
56. In which one of the following molecules strongest back donation of an electron pair halide to boron is expected? [JEE-Main-Online 2021]
 (A) BBr_3 (B) BI_3 (C) BCl_3 (D) BF_3
57. The spin-only magnetic moment value of B_2^+ species is _____ $\times 10^{-2}$ BM. [JEE-Main-Online 2021]
 (Nearest integer) [Given : $\sqrt{3} = 1.73$]
58. AB_3 is an interhalogen T-shaped molecule. The number of lone pairs of electrons on A is _____. (Integer answer) [JEE-Main-Online 2021]
59. The number of species having non-pyramidal shape among the following is _____. [JEE-Main-Online 2021]
 (i) SO_3 (ii) NO_3 (iii) PCl_3 (iv) CO_3^{2-}
60. According to molecular orbital theory, the number of unpaired electron(s) in O_2^{2-} is _____. [JEE-Main-Online 2021]
61. The difference between bond orders of CO and NO^+ is $\frac{x}{2}$ where $x =$ _____ (Round off to the Nearest integer) [JEE-Main-Online 2021]

62. The total number of electrons in all bonding molecular orbitals of O_2^{2-} is _____.
(Round off to the Nearest Integer). [JEE-Main-Online 2021]
63. The number of lone pairs of electrons on the central I atom in I_3^- is _____. [JEE-Main-Online 2021]
64. The number of sigma bonds in

$$\begin{array}{c} \text{H}_3\text{C}-\text{C}=\text{CH}-\text{C}\equiv\text{C}-\text{H} \\ | \\ \text{H} \end{array}$$
is _____. [JEE-Main-Online 2021]
65. The number of species below that have two lone pairs of electrons in their central atom is . (Round off to the Nearest Integer).
 SF_4 , BF_4^- , ClF_3 , AsF_3 , PCl_3 , BrF_5 , XeF_4 , SF_6 [JEE-Main-Online 2021]
66. Identify the **incorrect** statement for PCl_5 from the following. [JEE-Main-Online 2022]
 (A) In this molecule, orbitals of phosphorous are assumed to undergo sp^3d hybridization.
 (B) The geometry of PCl_5 is trigonal bipyramidal.
 (C) PCl has two axial bonds stronger than three equatorial bonds.
 (D) The three equatorial bonds of PCl_5 lie in a plane
67. The oxide which contains an odd electron at the nitrogen atom is : [JEE-Main-Online 2022]
 (A) N_2O (B) NO_2 (C) N_2O_3 (D) N_2O_5
68. Nitrogen gas is obtained by thermal decomposition of [JEE-Main-Online 2022]
 (A) $\text{Ba}(\text{NO}_3)_2$ (B) $\text{Ba}(\text{N}_3)_2$ (C) NaNO_2 (D) NaNO_3
69. Consider the ions/molecules [JEE-Main-Online 2022]
 O_2^+ , O_2 , O_2^- , O_2^{2-}
 For increasing bond order the **correct** option is :
 (A) $\text{O}_2^{2-} < \text{O}_2^- < \text{O}_2 < \text{O}_2^+$ (B) $\text{O}_2^- < \text{O}_2^{2-} < \text{O}_2 < \text{O}_2^+$
 (C) $\text{O}_2^- < \text{O}_2^{2-} < \text{O}_2^+ < \text{O}_2$ (D) $\text{O}_2^- < \text{O}_2^+ < \text{O}_2^{2-} < \text{O}_2$
70. Choose the **correct** stability order of group 13 elements in their +1 oxidation state. [JEE-Main-Online 2022]
 (A) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$ (B) $\text{Tl} < \text{In} < \text{Ga} < \text{Al}$ (C) $\text{Al} < \text{Ga} < \text{Tl} < \text{In}$ (D) $\text{Al} < \text{Tl} < \text{Ga} < \text{In}$
71. Bonding in which of the following diatomic molecule(s) become(s) stronger, on the basis of MO Theory, by removal of an electron?
 (A) NO (B) N_2 (C) O_2 (D) C_2 (E) B_2
 Choose the **most appropriate** answer from the options given below : [JEE-Main-Online 2022]
 (A) (A), (B), (C) only (B) (B), (C), (E) only (C) (A), (C) only (D) (D) only
72. Number of electron deficient molecules among the following
 PH_3 , B_2H_6 , CCl_4 , NH_3 , LiH and BCl_3 is [JEE-Main-Online 2022]
 (A) 0 (B) 1 (C) 2 (D) 3

73. The **correct** order of bond orders of C_2^{2-} , N_2^{2-} , O_2^{2-} is, respectively : [JEE-Main-Online 2022]
 (A) $C_2^{2-} < N_2^{2-} < O_2^{2-}$ (B) $O_2^{2-} < N_2^{2-} < C_2^{2-}$
 (C) $C_2^{2-} < O_2^{2-} < N_2^{2-}$ (D) $N_2^{2-} < C_2^{2-} < O_2^{2-}$
74. Arrange the following in the decreasing order of their covalent character : [JEE-Main-Online 2022]
 (a) LiCl (b) NaCl (c) KCl (d) CsCl
 Choose the most appropriate answer from the options given below :
 (A) (a) > (c) > (b) > (d) (B) (b) > (a) > (c) > (d)
 (C) (a) > (b) > (c) > (d) (D) (a) > (b) > (d) > (c)
75. Number of lone pair(s) of electrons on central atom and the shape of BrF_3 molecule respectively, are [JEE-Main-Online 2022]
 (A) 0, triangular planar (B) 1, pyramidal (C) 2, bent T-shape (D) 1, bent T-shape
76. PCl_5 is well known, but NCl_5 is not. Because, [JEE-Main-Online 2022]
 (A) Nitrogen is less reactive than phosphorous
 (B) Nitrogen doesn't have d-orbitals in its valence shell
 (C) Catenation tendency is weaker in nitrogen than phosphorous
 (D) Size of phosphorous is larger than nitrogen
77. The **correct** order of increasing intermolecular hydrogen bond strength is [JEE-Main-Online 2022]
 (A) $HCN < H_2O < NH_3$ (B) $HCN < CH_4 < NH_3$
 (C) $CH_4 < HCN < NH_3$ (D) $CH_4 < NH_3 < HCN$
78. The number of bridged oxygen atoms present in compound B formed from the following reactions is [JEE-Main-Online 2022]

$$Pb(NO_3)_2 \xrightarrow{673K} A + PbO + O_2$$

$$A \xrightarrow{\text{Dimerise}} B$$

 (A) 0 (B) 1 (C) 2 (D) 3
79. Consider the species CH_4 , NH_4^+ and BH_4^- . Choose the **correct** option with respect to these species. [JEE-Main-Online 2022]
 (A) They are isoelectronic and only two have tetrahedral structures
 (B) They are isoelectronic and all have tetrahedral structures
 (C) Only two are isoelectronic and all have tetrahedral structures
 (D) Only two are isoelectronic and only two have tetrahedral structures
80. Given below are the oxides : Na_2O , As_2O_3 , N_2O , NO and Cl_2O_7
 Number of amphoteric oxides is : [JEE-Main-Online 2022]
 (A) 0 (B) 1 (C) 2 (D) 3
81. The **correct** order of melting points of hydrides of group 16 elements is : [JEE-Main-Online 2022]
 (A) $H_2S < H_2Se < H_2Te < H_2O$ (B) $H_2O < H_2S < H_2Se < H_2Te$
 (C) $H_2S < H_2Te < H_2Se < H_2O$ (D) $H_2Se < H_2S < H_2Te < H_2O$

82. Identify the **correct** statement for B_2H_6 from those given below : **[JEE-Main-Online 2022]**
 (A) In B_2H_6 , all B-H bonds are equivalent. (B) In B_2H_6 , there are four 3-centre-2-electron bonds.
 (C) B_2H_6 is a Lewis acid. (D) B_2H_6 can be synthesized from both BF_3 and $NaBH_4$.
 (E) B_2H_6 is a planar molecule.
 Choose the most appropriate answer from the options given below : **[JEE-Main-Online 2022]**
 (A) (A) and (E) only (B) (B), (C) and (E) only
 (C) (C) and (D) only (D) (C) and (E) only
83. The most stable trihalide of nitrogen is : **[JEE-Main-Online 2022]**
 (A) NF_3 (B) NCl_3 (C) NBr_3 (D) NI_3
84. Among the following, basic oxide is : **[JEE-Main-Online 2022]**
 (A) SO_3 (B) SiO_2 (C) CaO (D) Al_2O_3
85. Among the given oxides of nitrogen ; N_2O , N_2O_3 , N_2O_4 and N_2O_5 , the number of compound/(s) having N-N bond is : **[JEE-Main-Online 2022]**
 (A) 1 (B) 2 (C) 3 (D) 4
86. Which of the following oxoacids of sulphur contains "S" in two different oxidation states? **[JEE-Main-Online 2022]**
 (A) $H_2S_2O_3$ (B) $H_2S_2O_6$ (C) $H_2S_2O_7$ (D) $H_2S_2O_8$
87. In the structure of SF_4 , the lone pair of electrons on S is in. **[JEE-Main-Online 2022]**
 (A) Equatorial position and there are two lone pair-bond pair repulsions at 90°
 (B) Equatorial position and there are three lone pair-bond pair repulsions at 90°
 (C) Axial position and there are three lone pair-bond pair repulsion at 90°
 (D) Axial position and there are two lone pair-bond pair repulsion at 90°
88. Given below are two statements : **[JEE-Main-Online 2022]**
Statement I : The pentavalent oxide of group-15 element, E_2O_5 , is less acidic than trivalent oxide, E_2O_3 , of the same element.
Statement II : The acidic character of trivalent oxide of group 15 elements, E_2O_3 , decreases down the group.
 In light of the above statements, choose **most appropriate** answer from the options given below :
 (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I true, but Statement II is false
 (D) Statement I false, but Statement II is true
89. Based upon VSEPR theory, match the shape (geometry) of the molecules in **List-I** with the molecules in **List-II** and select the most **appropriate** option. **[JEE-Main-Online 2022]**
- | List-I (Shape) | List-II (Molecules) |
|--|--|
| (A) T-shaped | (I) XeF_4 |
| (B) Trigonal planar | (II) SF_4 |
| (C) Square planar | (III) ClF_3 |
| (D) See-saw | (IV) BF_3 |
| (A) (A) – (I), (B) – (II), (C) – (III), (D) – (IV) | (B) (A) – (III), (B) – (IV), (C) – (I), (D) – (II) |
| (C) (A) – (III), (B) – (IV), (C) – (II), (D) – (I) | (D) (A) – (IV), (B) – (III), (C) – (I), (D) – (II) |

90. Match List-I with List-II :

[JEE-Main-Online 2022]

List-I (Oxide)	List-II (Nature)
(A) Cl_2O_7	(I) Amphoteric
(B) Na_2O	(II) Basic
(C) Al_2O_3	(III) Neutral
(D) N_2O	(IV) Acidic

Choose the **correct** answer from the options given below :

- (A) A-IV, B-III, C-I, D-II (B) A-IV, B-II, C-I, D-III
(C) A-II, B-IV, C-III, D-I (D) A-I, B-II, C-III, D-IV

91. Match List-I with List-II :

List-I (molecule)	List-II (hybridization; shape)
A. XeO_3	I. sp^3d ; linear
B. XeF_2	II. sp^3 ; pyramidal
C. XeOF_4	III. sp^3d^3 ; distorted octahedral
D. XeF_6	IV. sp^3d^2 ;square pyramidal

Choose the correct answer from the options given below :

[JEE-Main-Online 2022]

- (A) A-II, B-I, C-IV, D-III (B) A-II, B-IV, C-III, D-I
(C) A-IV, B-II, C-III, D-I (D) A-IV, B-II, C-I, D-III

92. Match List-I with List-II.

List-I (Compound)	List-II (Shape)
(A) BrF_5	(I) bent
(B) $[\text{CrF}_6]^{3-}$	(II) square pyramidal
(C) O_3	(III) trigonal bipyramidal
(D) PCl_5	(IV) octahedral

Choose the correct answer from the options given below :

[JEE-Main-Online 2022]

- (A) (A) - (I), (B) - (II), (C) - (III), (D) - (IV) (B) (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
(C) (A) - (II), (B) - (IV), (C) - (I), (D) - (III) (D) (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

93. Match List-I with List-II

[JEE-Main-Online 2022]

List-I	List-II
(A) $\Psi_{\text{MO}} = \Psi_{\text{A}} - \Psi_{\text{B}}$	(I) Dipole moment
(B) $\mu = Q \times r$	(II) Bonding molecular orbital
(C) $\frac{N_{\text{b}} - N_{\text{a}}}{2}$	(III) Anti-bonding molecular orbital
(D) $\Psi_{\text{MO}} = \Psi_{\text{A}} + \Psi_{\text{B}}$	(IV) Bond order
(A) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)	(B) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
(C) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)	(D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

94. Given below are two statements :

Statement-I : The chlorides of Be and Al have Cl-bridged structure. Both are soluble in organic solvents and act as Lewis bases.

Statement-II : Hydroxides of Be and Al dissolve in excess alkali to give beryllate and aluminate ions.

In the light of the above statements. Choose the correct answer from the options given below.

[JEE-Main-Online 2022]

- (A) Both statement I and Statement II are true (B) Both statement I and Statement II are false
(C) Statement I is true but Statement II is false (D) Statement I is false but Statement II is true

95. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**
Assertion A : Zero orbital overlap is an out of phase overlap.
Reason : It results due to different orientation/direction of approach of orbitals.
 In the light of the above statements. Choose the correct answer from the options given below
[JEE-Main-Online 2022]
- (A) Both A and R are true and R is the correct explanation of A
 (B) Both A and R are true but R is NOT the correct explanation of A
 (C) A is true but R is false
 (D) A is false but R is true
96. Given below are two statements.
Statement-I : Stannane is an example of a molecular hydride.
Statement-II : Stannane is a planar molecule.
 In the light of the above statement, choose the most appropriate answer from the options given below :
[JEE-Main-Online 2022]
- (A) Both Statement I and Statement II are true. (B) Both Statement I and Statement II are false.
 (C) Statement I is true but Statement II is false. (D) Statement I is false but Statement II is true.
97. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : Boric acid is a weak acid
Reason R : Boric acid is not able to release H^+ ion on its own. It receives OH^- ion from water and releases H^+ ion.
 In the light of the above statements, choose the most appropriate answer from the options given below.
[JEE-Main-Online 2022]
- (A) Both A and R are correct and R is the correct explanation of A
 (B) Both A and R are correct but R is NOT the correct explanation of A
 (C) A is correct but R is not correct
 (D) A is not correct but R is correct
98. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**
Assertion (A) : Boron is unable to form BF_6^{3-}
Reason (R) : Size of B is very small.
 In the light of the above statements, choose the correct answer from the options given below :
[JEE-Main-Online 2022]
- (A) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (B) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (C) (A) is true but (R) is false
 (D) (A) is false but (R) is true
99. Given below are two statements.
Statement-I : O_2 , Cu^{2+} and Fe^{3+} are weakly attracted by magnetic field and are magnetized in the same direction as magnetic field.
Statement-II : $NaCl$ and H_2O are weakly magnetized in opposite direction to magnetic field.
 In the light of the above statements, choose the most appropriate answer from the options given below :
[JEE-Main-Online 2022]
- (A) Both Statement I and Statement II are correct.
 (B) Both Statement I and Statement II are incorrect.
 (C) Statement I is correct but Statement II is incorrect.
 (D) Statement I is incorrect but Statement II is correct.

100. The total number of Mn = O bonds in Mn_2O_7 is _____. [JEE-Main-Online 2022]
 (A) 4 (B) 5 (C) 6 (D) 3
101. Number of lone pairs of electrons in the central atom of SCl_2 , O_3 , ClF_3 and SF_6 , respectively, are : [JEE-Main-Online 2022]
 (A) 0, 1, 2 and 2 (B) 2, 1, 2 and 0 (C) 1, 2, 2 and 0 (D) 2, 1, 2 and 0
102. Which oxoacid of phosphorous has the highest number of oxygen atoms present in its chemical formula? [JEE-Main-Online 2022]
 (A) Pyrophosphorous acid (B) Hypophosphoric acid
 (C) Phosphoric acid (D) Pyrophosphoric acid
103. Arrange the following in increasing order of their covalent character.
 (A) CaF_2 (B) CaCl_2 (C) CaBr_2 (D) CaI_2
 Choose the correct answer from the options given below. [JEE-Main-Online 2022]
 (A) $B < A < C < D$ (B) $A < B < C < D$
 (C) $A < B < D < C$ (D) $A < C < B < D$
104. The total number of acidic oxides from the following list is : NO , N_2O , B_2O_3 , N_2O_5 , CO , SO_3 , P_4O_{10} [JEE-Main-Online 2022]
 (A) 3 (B) 4 (C) 5 (D) 6
105. Which of the following pair of molecules contain odd electron molecule and an expanded octet molecule? [JEE-Main-Online 2022]
 (A) BCl_3 and SF_6 (B) NO and H_2SO_4 (C) SF_6 and H_2SO_4 (D) BCl_3 and NO
106. The sum of number of lone pairs of electrons present on the central atoms of XeO_3 , XeOF_4 and XeF_6 is _____. [JEE-Main-Online 2022]
107. According to MO theory, number of species/ions from the following having identical bond order is ____: [JEE-Main-Online 2022]
 CN^- , NO^+ , O_2 , O_2^+ , O_2^{2+}
108. Among the following species
 N_2 , N_2^+ , N_2^- , N_2^{2-} , O_2 , O_2^+ , O_2^- , O_2^{2-}
 the number of species showing diamagnetism is [JEE-Main-Online 2022]
109. Consider, PF_5 , BrF_5 , PCl_3 , SF_6 , $[\text{ICl}_4]^-$, ClF_3 and IF_5 .
 Amongst the above molecule(s)/ion(s), the number of molecule(s)/ion(s) having sp^3d^2 hybridisation is _____. [JEE-Main-Online 2022]
110. Consider the following sulphur based oxoacids.
 H_2SO_3 , H_2SO_4 , $\text{H}_2\text{S}_2\text{O}_8$ and $\text{H}_2\text{S}_2\text{O}_7$.
 Amongst these oxoacids, the number of those with peroxo(O-O) bond is _____. [JEE-Main-Online 2022]
111. Amongst the following the number of oxide(s) which are paramagnetic in nature is
 Na_2O , KO_2 , NO_2 , N_2O , ClO_2 , NO , SO_2 , Cl_2O [JEE-Main-Online 2022]
112. The number of paramagnetic species among the following is _____. [JEE-Main-Online 2022]
 B_2 , Li_2 , C_2 , C_2^- , O_2^{2-} , O_2^+ and He_2^+
113. The number of interhalogens from the following having square pyramidal structure is :
 ClF_3 , IF_7 , BrF_5 , BrF_3 , I_2Cl_6 , IF_5 , ClF , ClF_5 [JEE-Main-Online 2022]

114. The number of molecule(s) or ion(s) from the following having non-planar structure is _____.
 NO_3^- , H_2O_2 , BF_3 , PCl_3 , XeF_4 , SF_4 , XeO_3 , PH_4^+ , SO_3 , $[\text{Al}(\text{OH})_4]^-$ [JEE-Main-Online 2022]
115. Amongst BeF_2 , BF_3 , H_2O , NH_3 , CCl_4 and HCl , the number of molecules with non-zero net dipole moment is _____. [JEE-Main-Online 2022]
116. The hybridization of P exhibited in PF_5 is sp_xd_y . The value of y is _____. [JEE-Main-Online 2022]
117. The difference in oxidation state of chromium in chromate and dichromate salts is _____. [JEE-Main-Online 2022]
118. Amongst SF_4 , XeF_4 , CF_4 and H_2O , the number of species with two lone pair of electrons is _____. [JEE-Main-Online 2022]
119. Resonance in carbonate ion (CO_3^{2-}) is [JEE-Main-Online 2023]
-
- Which of the following is true?
- (A) It is possible to identify each structure individually by some physical or chemical method.
 (B) All these structures are in dynamic equilibrium with each other.
 (C) CO_3^{2-} has a single structure i.e., resonance hybrid of the above three structures.
 (D) Each structure exists for equal amount of time.
120. The Lewis acid character of boron tri halides follows the order ; [JEE-Main-Online 2023]
- (A) $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3 > \text{BO}_3$ (B) $\text{BCl}_3 > \text{BF}_3 > \text{BBr}_3 > \text{BI}_3$
 (C) $\text{BI}_3 > \text{BBr}_3 > \text{BCl}_3 > \text{BF}_3$ (D) $\text{BBr}_3 > \text{BI}_3 > \text{BCl}_3 > \text{BF}_3$
121. Match List-I with List-II
- | List-I | List-II |
|---------------------|---------------------|
| (A) XeF_4 | (I) See-saw |
| (B) SF_4 | (II) Square planar |
| (C) NH_4^+ | (III) Bent T-shaped |
| (D) BrF_3 | (IV) Tetrahedral |
- Choose the correct answer from the options given below : [JEE-Main-Online 2023]
- (A) A-II B-I C-III D-IV
 (B) A-IV B-I C-II D-III
 (C) A-II B-I C-IV D-III
 (D) A-IV B-III C-II D-I
122. Boric acid is solid, whereas BF_3 is gas at room temperature because of [JEE-Main-Online 2023]
- (A) Strong van der Waal's interaction in Boric acid (B) Strong hydrogen bond in Boric acid
 (C) Strong ionic bond in Boric acid (D) Strong covalent bond in BF_3
123. Bond dissociation energy of "E-H" bond of the " H_2E " hydrides of group 16 elements (given below), follows order.
- (A) O (B) S (C) Se (D) Te
- Choose the correct from the options given below : [JEE-Main-Online 2023]
- (A) $A > B > C > D$ (B) $D > C > B > A$
 (C) $A > B > D > C$ (D) $B > A > C > D$

124. The correct order of hydration enthalpies is :
 (A) K^+ (B) Rb^+ (C) Mg^{2+} (D) Cs^+ (E) Ca^{2+}
 Choose the correct answer from the options given below : [JEE-Main-Online 2023]
 (A) $C > E > A > D > B$ (B) $C > A > E > B > D$ (C) $C > E > A > B > D$ (D) $E > C > A > B > D$
125. Match List-I with List-II

List-I (molecules/ions)	List-II (No. of lone pairs of e^- on central atom)
(A) IF_7	(I) Three
(B) ICl_4^-	(II) One
(C) XeF_6	(III) Two
(D) XeF_2	(IV) Zero

 Choose the correct answer from the options given below : [JEE-Main-Online 2023]
 (A) A-IV B-I C-II D-III
 (B) A-II B-III C-IV D-I
 (C) A-IV B-III C-II D-I
 (D) A-II B-I C-IV D-III
126. For OF_2 molecule consider the following :
 (A) Number of lone pairs on oxygen is 2. (B) FOF angle is less than 104.5° .
 (C) Molecule is bent 'V' shaped. (D) Molecular geometry is linear.
 Correct options are : [JEE-Main-Online 2023]
 (A) A, B, C only (B) A, B, D only (C) C, D, E only (D) A, C, D only
127. The bond dissociation energy is highest for [JEE-Main-Online 2023]
 (A) Br_2 (B) F_2 (C) I_2 (D) Cl_2
128. What is the number of unpaired electron (s) in the highest occupied molecular orbital of the following species : N_2 ; N_2^+ ; O_2 ; O_2^+ ? [JEE-Main-Online 2023]
 (A) 0, 1, 0, 1 (B) 2, 1, 2, 1 (C) 0, 1, 2, 1 (D) 2, 1, 0, 1
129. The correct order of bond enthalpy (kJ mol^{-1}) is : [JEE-Main-Online 2023]
 (A) $C - C > Si - Si > Ge - Ge > Sn - Sn$ (B) $C - C > Si - Si > Sn - Sn > Ge - Ge$
 (C) $Si - Si > C - C > Sn - Sn > Ge - Ge$ (D) $Si - Si > C - C > Ge - Ge > Sn - Sn$
130. The magnetic behavior of Li_2O , Na_2O_2 and KO_2 , respectively, are [JEE-Main-Online 2023]
 (A) paramagnetic, diamagnetic and paramagnetic
 (B) diamagnetic, diamagnetic and paramagnetic
 (C) diamagnetic, paramagnetic and diamagnetic
 (D) paramagnetic, paramagnetic and diamagnetic
131. O – O bond length in H_2O_2 is $\underline{X}$ than the O – O bond length in F_2O_2 . The O – H bond length in H_2O_2 is $\underline{Y}$ than that of the O – F bond in F_2O_2 .
 Choose the correct option for $\underline{X}$ and $\underline{Y}$ from those given below : [JEE-Main-Online 2023]
 (A) X - longer, Y - longer (B) X - longer, Y - shorter
 (C) X - shorter, Y - shorter (D) X - shorter, Y - longer
132. Order of Covalent bond :
 (A) $KF > KI$; $LiF > KF$ (B) $KF < KI$; $LiF > KF$
 (C) $SnCl_4 > SnCl_2$; $CuCl > NaCl$ (D) $LiF > KF$; $CuCl < NaCl$
 (E) $KF < KI$; $CuCl > NaCl$
 Choose the correct answer from the options given below : [JEE-Main-Online 2023]
 (A) B, C only (B) C, E only (C) B, C, E only (D) A, B only

133. Which of the Phosphorus oxoacid can create silver mirror from AgNO_3 solution? [JEE-Main-Online 2023]
 (A) $(\text{HPO}_3)_n$ (B) $\text{H}_4\text{P}_2\text{O}_7$ (C) $\text{H}_4\text{P}_2\text{O}_6$ (D) $\text{H}_4\text{P}_2\text{O}_5$
134. **Statement I** : Dipole moment is a vector quantity and by convention it is depicted by a small arrow with tail on the negative centre and head pointing towards the positive centre.
Statement II : The crossed arrow of the dipole moment symbolizes the direction of the shift of charges in the molecules.
 In the light of the above statements, choose the most appropriate answer from the options given below. [JEE-Main-Online 2023]
 (A) Both Statement-I and Statement-II are correct
 (B) Both Statement-I and Statement-II are incorrect
 (C) Statement-I is correct but Statement-II is incorrect
 (D) Statement-I is incorrect but Statement-II is correct
135. According to MO theory the bond orders for O_2^{2-} , CO and NO^+ respectively, are [JEE-Main-Online 2023]
 (A) 1, 2 and 3 (B) 1, 3 and 2 (C) 1, 3 and 3 (D) 2, 3 and 3
136. The alkaline earth metal sulphate(s) which are readily soluble in water is/are :
 (A) BeSO_4 (B) MgSO_4 (C) CaSO_4 (D) SrSO_4
 (E) BaSO_4
 Choose the **correct** answer from the options given below : [JEE-Main-Online 2023]
 (A) B only (B) B and C (C) A and B (D) A only
137. For compound having the formula $\text{Ga}[\text{AlCl}_4]$, the correct option from the following is [JEE-Main-Online 2023]
 (A) Cl forms bond with both Al and Ga in $\text{Ga}[\text{AlCl}_4]$
 (B) Oxidation state of Ga in the salt GaAlCl_4 is +3.
 (C) Ga is more electronegative than Al and is present as a cationic part of the salt GaAlCl_4
 (D) Ga is coordinated with Cl in GaAlCl_4
138. Match List-I with List-II :

List-I (Oxide)	List-II (Type of bond)
(A) N_2O_4	(I) 1 N = O bond
(B) NO_2	(II) 1 N–O–N bond
(C) N_2O_5	(III) 1 N–N bond
(D) N_2O	(IV) 1 N=N / N=N bond

 Choose the correct answer from the options given below : [JEE-Main-Online 2023]
 (A) A – III, B – I, C – II, D – IV (B) A – III, B – I, C – IV, D – II
 (C) A – II, B – I, C – III, D – IV (D) A – II, B – IV, C – III, D – I
139. Match List-I with List-II :

List-I (Species)	List-II (Geometry / Shape)
A. H_3O^+	I. Tetrahedral
B. Acetylide anion	II. Linear
C. NH_4^+	III. Pyramidal
D. ClO_2^-	IV. Bent

 Choose the correct answer from the options given below : [JEE-Main-Online 2023]
 (A) A-III, B-IV, C-II, D-I (B) A-III, B-IV, C-I, D-II
 (C) A-III, B-II, C-I, D-IV (D) A-III, B-I, C-II, D-IV

140. Given below are two statements :
Statement-I : SO_2 and H_2O both possess V-shaped structure.
Statement-II : The bond angle of SO_2 is less than that of H_2O .
 In the light of the above statements, choose the most appropriate answer from the options given below:
[JEE-Main-Online 2023]
- (A) Statement I is incorrect but Statement II is correct
 (B) Both Statement I and Statement II are correct
 (C) Statement I is correct but Statement II is incorrect
 (D) Both Statement I and Statement II are incorrect
141. The covalency and oxidation state of boron in $[\text{BF}_4]^-$, are
[JEE-Main-Online 2023]
- (A) 3 and 5 (B) 4 and 3 (C) 3 and 4 (D) 4 and 4
142. In which of the following processes, the bond order increases and paramagnetic character changes to diamagnetic one?
[JEE-Main-Online 2023]
- (A) $\text{O}_2 \rightarrow \text{O}_2^+$ (B) $\text{N}_2 \rightarrow \text{N}_2^+$ (C) $\text{O}_2 \rightarrow \text{O}_2^{2-}$ (D) $\text{NO} \rightarrow \text{NO}^+$
143. The number of P – O – P bonds in $\text{H}_4\text{P}_2\text{O}_7$, $(\text{HPO}_3)_3$ and P_4O_{10} are respectively **[JEE-Main-Online 2023]**
- (A) 1, 3, 6 (B) 0, 3, 4 (C) 0, 3, 6 (D) 1, 2, 4
144. Consider the following statements:
 (A) NF_3 molecule has a trigonal planar structure.
 (B) Bond length of N_2 is shorter than O_2 .
 (C) Isoelectronic molecules or ions have identical bond order.
 (D) Dipole moment of H_2S is higher than that of water molecule.
 Choose the correct answer from the options given below:
[JEE-Main-Online 2023]
- (A) (C) and (D) are correct (B) (B) and (C) are correct
 (C) (A) and (B) are correct (D) (A) and (D) are correct
145. Group-13 elements react with O_2 in amorphous form to form oxides of type M_2O_3 (M = element). Which among the following is the most basic oxide?
[JEE-Main-Online 2023]
- (A) B_2O_3 (B) Al_2O_3 (C) Ti_2O_3 (D) Ga_2O_3
146. The bond order and magnetic property of acetylide ion are same as that of **[JEE-Main-Online 2023]**
- (A) O_2^- (B) O_2^+ (C) N_2^+ (D) NO^+
147. Given below are two statements :
Statement-I : SbCl_3 is more covalent than SbCl_5
Statement-II : The higher oxides of halogens also tend to be more stable than the lower ones.
 In the light of the above statements, choose the most appropriate answer from the options given below
[JEE-Main-Online 2023]
- (A) Both Statement I and Statement II are incorrect
 (B) Both statement I and Statement II are correct
 (C) Statement I is incorrect but Statement II is correct
 (D) Statement I is correct but Statement II is incorrect
148. Structures of BeCl_2 in solid state, vapour phase and at very high temperature respectively are :
[JEE-Main-Online 2023]
- (A) Polymeric, Dimeric, Monomeric (B) Dimeric, Polymeric, Monomeric
 (C) Polymeric, Monomeric, Dimeric (D) Monomeric, Dimeric, Polymeric

149. Which one of the following pairs is an example of polar molecular solids? [JEE-Main-Online 2023]
 (A) HCl (s), AlN (s) (B) MgO (s), SO₂(s)
 (C) SO₂(s), NH₃(s) (D) SO₂(s), CO₂(s)
150. Arrange the following gases in increasing order of van der Waals constant 'a'
 A. Ar B. CH₄ C. H₂O D. C₆H₆
 Choose the correct option from the following. [JEE-Main-Online 2023]
 (A) B, C, D and A (B) D, C, B and A
 (C) A, B, C and D (D) C, D, B and A
151. The ratio of sigma and π bonds present in pyrophosphoric acid is _____. [JEE-Main-Online 2023]
152. The total number of isoelectronic species from the given set is _____.
 O²⁻, F⁻, Al, Mg²⁺, Na⁺, O⁺, Mg, Al³⁺, F [JEE-Main-Online 2023]
153. The maximum number of lone pairs of electron on the central atom from the following species is _____.
 ClO₃⁻, XeF₄, SF₄ and I₃⁻ [JEE-Main-Online 2023]
154. If the formula of Borax is Na₂B₄O_x(OH)_y·zH₂O, then x + y + z = _____. [JEE-Main-Online 2023]
155. The number of bent-shaped molecule/s from the following is _____.
 N₃⁻, NO₂⁻, I₃⁻, O₃, SO₂ [JEE-Main-Online 2023]
156. The number of following factors which affect the percent covalent character of the ionic bond is _____.
 (i) Polarising power of cation (ii) Extent of distortion of anion
 (iii) Polarisability of the anion (iv) Polarising power of anion [JEE-Main-Online 2023]
157. The number of species having a square planar shape from the following is _____.
 XeF₄, SF₄, SiF₄, BF₄⁻, BrF₄⁻, [Cu(NH₃)₄]²⁺, [FeCl₄]²⁻, [PtCl₄]²⁻ [JEE-Main-Online 2023]
158. The sum of lone pairs present on the central atom of the interhalogen IF₅ and IF₇ is _____. [JEE-Main-Online 2023]
159. The number of species from the following which have square pyramidal structure is _____.
 PF₅, BrF₄⁻, IF₅, BrF₅, XeOF₄, ICl₄⁻ [JEE-Main-Online 2023]
160. The number of species from the following carrying a single lone pair on central atom Xenon is _____.
 XeF₅⁺, XeO₃, XeO₂F₂, XeF₅⁻, XeO₃F₂, XeOF₄, XeF₄ [JEE-Main-Online 2023]
161. In the following reactions, the total number of oxygen atoms in X and Y is _____.

$$\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{X}$$

$$\text{Cl}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2\text{Y}$$
 [JEE-Main-Online 2023]
162. The number of molecules from the following which contain only two lone pair of electrons is _____.
 H₂O, N₂, CO, XeF₄, NH₃, NO, CO₂, F₂ [JEE-Main-Online 2023]
163. Sum of oxidation states of bromine in bromic acid and perbromic acid is _____. [JEE-Main-Online 2023]
164. Sum of π -bonds present in peroxodisulphuric acid and pyrosulphuric acid is _____. [JEE-Main-Online 2023]
165. The number of molecules or ions from the following, which do not have odd number of electrons are _____.
 (A) NO₂ (B) ICl₄⁻ (C) BrF₃ (D) ClO₂
 (E) NO₂⁺ (F) NO [JEE-Main-Online 2023]

166. The total number of lone pairs of electrons on oxygen atoms of ozone is _____.
[JEE-Main-Online 2023]
167. On heating, LiNO_3 gives how many compounds among the following? _____.
 Li_2O , N_2 , O_2 , LiNO_2 , NO_2
[JEE-Main-Online 2023]
168. The number of given orbitals which have electron density along the axis is _____.
 $p_x, p_y, p_z, d_{xy}, d_{yz}, d_{xz}, d_{z^2}, d_{x^2-y^2}$
[JEE-Main-Online 2023]
169. Amongst the following, the number of species having the linear shape is _____.
 $\text{XeF}_2, \text{I}_3^+, \text{C}_3\text{O}_2, \text{I}_3^-, \text{CO}_2, \text{SO}_2, \text{BeCl}_2$ and $\text{BCl}_2^\ominus$
[JEE-Main-Online 2023]
170. Total number of acidic oxides among
 $\text{N}_2\text{O}_3, \text{NO}_2, \text{N}_2\text{O}, \text{Cl}_2\text{O}_7, \text{SO}_2, \text{CO}, \text{CaO}, \text{Na}_2\text{O}$ and NO is _____. [JEE-Main-Online 2023]
171. Element not showing variable oxidation state is : [JEE-Main-Online 2024]
(A) Chlorine (B) Fluorine (C) Bromine (D) Iodine
172. Consider the oxides of group 14 elements
 $\text{SiO}_2, \text{GeO}_2, \text{SnO}_2, \text{PbO}_2, \text{CO}$ and GeO . The amphoteric oxides are [JEE-Main-Online 2024]
(A) $\text{SnO}_2, \text{PbO}_2$ (B) GeO, GeO_2 (C) $\text{SiO}_2, \text{GeO}_2$ (D) SnO_2, CO
173. Which of the following is least ionic ? [JEE-Main-Online 2024]
(A) BaCl_2 (B) AgCl (C) KCl (D) CoCl_2
174. The linear combination of atomic orbitals to form molecular orbitals takes place only when the combining atomic orbitals
(a) have the same energy
(b) have the minimum overlap
(c) have same symmetry about the molecular axis
(d) have different symmetry about the molecular axis
Choose the **most appropriate** from the options given below : [JEE-Main-Online 2024]
(A) b and d only (B) a and c only (C) a, b, c only (D) b, c, d only
175. The difference in energy between the actual structure and the lowest energy resonance structure for the given compound is : [JEE-Main-Online 2024]
(A) Ionization energy (B) Electromeric energy
(C) Hyperconjugation energy (D) Resonance energy
176. Arrange the bonds in order of increasing ionic character in the molecules. $\text{LiF}, \text{K}_2\text{O}, \text{N}_2, \text{SO}_2$ and ClF_3 : [JEE-Main-Online 2024]
(A) $\text{LiF} < \text{K}_2\text{O} < \text{ClF}_3 < \text{SO}_2 < \text{N}_2$ (B) $\text{ClF}_3 < \text{N}_2 < \text{SO}_2 < \text{K}_2\text{O} < \text{LiF}$
(C) $\text{N}_2 < \text{SO}_2 < \text{ClF}_3 < \text{K}_2\text{O} < \text{LiF}$ (D) $\text{N}_2 < \text{ClF}_3 < \text{SO}_2 < \text{K}_2\text{O} < \text{LiF}$
177. Choose the correct statements about the hydrides of group 15 elements.
(I) The stability of the hydrides decreases in the order $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3$
(II) The reducing ability of the hydride increases in the order $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3 < \text{BiH}_3$
(III) Among the hydrides, NH_3 is strong reducing agent while BiH_3 is mild reducing agent
(IV) The basicity of the hydrides increases in the order $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3 < \text{BiH}_3$
Choose the most appropriate from the options given below : [JEE-Main-Online 2024]
(A) I and IV only (B) I and II only (C) II and III only (D) III and IV only

178. Given below are two statements : One is labelled as **Assertion A** and the other is labelled as **Reason R**:
Assertion A : H_2Te is more acidic than H_2S .
Reason R : Bond dissociation enthalpy of H_2Te is lower than H_2S .
 In the light of the above statements, choose the *most appropriate* from the options given below :
[JEE-Main-Online 2024]
- (A) A is false but R is true
 (B) Both A and R are true but R is NOT the correct explanation of A.
 (C) A is true but R is false
 (D) Both A and R are true and R is the correct explanation of A
179. Given below are two statements :
Statement - I : Since Fluorine is more electronegative than nitrogen, the net dipole moment of NF_3 is greater than NH_3 .
Statement - II : In NH_3 , the orbital dipole due to lone pair and the dipole moment of NH bonds are in opposite direction, but in NF_3 the orbital dipole due to lone pair and dipole moments of N – F bonds are in same direction.
 In the light of the above statements, choose the *most appropriate* from the options given below :
[JEE-Main-Online 2024]
- (A) Both Statement I and Statement - II are false
 (B) Both Statement I and Statement - II are true
 (C) Statement I is false but Statement - II is true
 (D) Statement I is true but Statement - II is false
180. Given below are two statements :
Statement I : Noble gases have very high boiling points
Statement II : Noble gases are monoatomic gases. They are held together by strong dispersion forces. Because of this they are liquefied at very low temperature. Hence, they have very high boiling points.
 In the light of the above statements, choose the *correct answer* from the options given below :
[JEE-Main-Online 2024]
- (A) Both Statement I and Statement II are false
 (B) Both Statement I and Statement II are true
 (C) Statement I is true but Statement II is false
 (D) Statement I is false but Statement II is true
181. Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : PH_3 has lower boiling point than NH_3 .
Reason (R) : In liquid state NH_3 molecules are associated through vander Waal's forces, but PH_3 molecules are associated through hydrogen bonding.
 In the light of the above statements, choose the most appropriate answer from the options given below :
[JEE-Main-Online 2024]
- (A) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 (B) (A) is not correct but (R) is correct
 (C) Both (A) and (R) are correct but (R) is the correct explanation of (A)
 (D) (A) is correct but (R) is not correct

182. Given below are two statements :

Statement (I) : A π bonding MO has lower electron density above and below the inter-nuclear axis.

Statement (II) : The π^* antibonding MO has a node between the nuclei.

In the light of the above statements, choose the most appropriate answer from the options given below :

[JEE-Main-Online 2024]

- (A) Both Statement I and Statement II are true
(B) Statement I is true but Statement II is false
(C) Statement I is false but Statement II is true
(D) Both Statement I and Statement II are false

183. Match List - I with List - II

List - I
Molecule

- (A) BrF_5
(B) H_2O
(C) ClF_3
(D) SF_4

List - II
Shape

- (I) T-shape
(II) See saw
(III) Bent
(IV) Square pyramidal

Choose the correct answer from the options given below :

[JEE-Main-Online 2024]

- (A) (A) – (IV), (B) – (III), (C) – (I), (D) – (II)
(B) (A) – (II), (B) – (I), (C) – (III), (D) – (IV)
(C) (A) – (III), (B) – (IV), (C) – (I), (D) – (II)
(D) (A) – (I), (B) – (II), (C) – (IV), (D) – (III)

184. The total number of molecules with zero dipole moment among CH_4 , BF_3 , H_2O , HF , NH_3 , CO_2 and SO_2 is _____.

[JEE-Main-Online 2024]

185. The number of species from the following in which the central atom uses sp^3 hybrid orbitals in its bonding is _____.

NH_3 , SO_2 , SiO_2 , BeCl_2 , CO_2 , H_2O , CH_4 , BF_3

[JEE-Main-Online 2024]

186. The total number of anti bonding molecular orbitals, formed from 2s and 2p atomic orbitals in a diatomic molecule is _____.

[JEE-Main-Online 2024]

187. The total number of molecular orbitals formed from 2s and 2p atomic orbitals of a diatomic molecule is _____.

[JEE-Main-Online 2024]

188. The number of species from the following which are paramagnetic and with bond order equal to one is _____.

H_2 , He_2^+ , O_2^+ , N_2^{2-} , O_2^{2-} , F_2 , Ne_2^+ , B_2

[JEE-Main-Online 2024]

189. Number of compounds with one lone pair of electrons on central atom amongst following is _____.

O_3 , H_2O , SF_4 , ClF_3 , NH_3 , BrF_5 , XeF_4

[JEE-Main-Online 2024]

190. Sum of bond order of CO and NO^+ is _____.

[JEE-Main-Online 2024]

191. From the given list, the number of compounds with +4 oxidation state of Sulphur is _____.

SO_3 , H_2SO_3 , SOCl_2 , SF_4 , BaSO_4 , $\text{H}_2\text{S}_2\text{O}_7$

[JEE-Main-Online 2024]

192. The number of non-polar molecules from the following is _____.

HF , H_2O , SO_2 , H_2 , CO_2 , CH_4 , NH_3 , HCl , CHCl_3 , BF_3

[JEE-Main-Online 2024]

193. The lowest oxidation number of an atom in a compound A_2B is -2 . The number of electrons in its valence shell is _____.

[JEE-Main-Online 2024]

194. A diatomic molecule has a dipole moment of 1.2 D. If the bond distance is 1 Å, then fractional charge on each atom is _____ $\times 10^{-1}$ esu.
(Given : 1 D = 10^{-18} esucm) [JEE-Main-Online 2024]
195. The shape of carbocation is : [JEE-Main-Online 2024]
(A) tetrahedral (B) trigonal planar (C) diagonal (D) diagonal pyramidal
196. The correct increasing order for bond angles among BF_3 , PF_3 and ClF_3 is: [JEE-Main-Online 2024]
(A) $\text{PF}_3 < \text{BF}_3 < \text{ClF}_3$ (B) $\text{BF}_3 < \text{PF}_3 < \text{ClF}_3$
(C) $\text{ClF}_3 < \text{PF}_3 < \text{BF}_3$ (D) $\text{BF}_3 = \text{PF}_3 < \text{ClF}_3$
197. The **incorrect** statement regarding ethyne is : [JEE-Main-Online 2024]
(A) Ethyne is linear
(B) The C – C bonds in ethyne is shorter than that in ethene
(C) The carbon – carbon bonds in ethyne is weaker than that in ethene
(D) Both carbons are sp hybridised
198. When Ψ_A and Ψ_B are the wave functions of atomic orbitals, then σ^* is represented by : [JEE-Main-Online 2024]
(A) $\Psi_A + 2\Psi_B$ (B) $\Psi_A - 2\Psi_B$ (C) $\Psi_A - \Psi_B$ (D) $\Psi_A + \Psi_B$
199. The number of species from the following that have pyramidal geometry around the central atom is _____ [JEE-Main-Online 2024]
 $\text{S}_2\text{O}_3^{2-}$, SO_4^{2-} , SO_3^{2-} , $\text{S}_2\text{O}_7^{2-}$
(A) 1 (B) 3 (C) 2 (D) 4
200. The correct statement/s about Hydrogen bonding is/are
(A) Hydrogen bonding exists when H is covalently bonded to the highly electronegative atom
(B) Intermolecular H bonding is present in o-nitro phenol
(C) Intramolecular H bonding is present in HF
(D) The magnitude of H bonding depends on the physical state of the compound
(E) H-bonding has powerful effect on the structure and properties of compounds
Choose the correct answer from the options given below : [JEE-Main-Online 2024]
(A) A only (B) A, D, E only (C) A, B, D only (D) A, B, C only
201. Which one of the following molecules has maximum dipole moment? [JEE-Main-Online 2024]
(A) CH_4 (B) NF_3 (C) NH_3 (D) PF_5
202. Number of molecules/ions from the following in which the central atom is involved in sp^3 hybridization is _____. [JEE-Main-Online 2024]
 NO_3^- , BCl_3 , ClO_2^- , ClO_3
(A) 3 (B) 4 (C) 2 (D) 1
203. In which one of the following pairs the central atoms exhibit sp^2 hybridization ? [JEE-Main-Online 2024]
(A) BF_3 and NO_2^- (B) NH_2^- and H_2O (C) NH_2^- and BF_3 (D) H_2O and NO_2
204. The statement(s) that are correct about the species O^{2-} , F^- , Na^+ and Mg^{2+} .
(A) All are isoelectronic (B) All have the same nuclear charge
(C) O^{2-} has the largest ionic radii (D) Mg^{2+} has the smallest ionic radii
Choose the most appropriate answer from the options given below : [JEE-Main-Online 2024]
(A) (A), (C) and (D) only (B) (B), (C) and (D) only
(C) (A), (B), (C) and (D) (D) (C) and (D) only

205. Number of σ and π bonds present in ethylene molecule is respectively : **[JEE-Main-Online 2024]**
 (A) 5 and 2 (B) 4 and 1 (C) 3 and 1 (D) 5 and 1
206. Given below are two statements:
Statement (I) : The oxidation state of an element in a particular compound is the charge acquired by its atom on the basis of electron gain enthalpy consideration from other atoms in the molecule.
Statement (II) : $p\pi - p\pi$ bond formation is more prevalent to second period elements over other periods.
 In the light of the above statements, choose the most appropriate answer from the options given below : **[JEE-Main-Online 2024]**
 (A) Both **Statement I** and **Statement II** are incorrect
 (B) Both **Statement I** and **Statement II** are correct
 (C) **Statement I** is correct but **Statement II** is incorrect
 (D) **Statement I** is incorrect but **Statement II** is correct
207. Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : NH_3 and NF_3 molecule have pyramidal shape with a lone pair of electrons on nitrogen atom. The resultant dipole moment of NH_3 is greater than that of NF_3 .
Reason (R) : In NH_3 , the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of the N – H bonds. F is the most electronegative element.
 In the light of the above statements, choose the correct answer from the options given below : **[JEE-Main-Online 2024]**
 (A) (A) is false but (R) is true
 (B) (A) is true but (R) is false
 (C) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
 (D) Both (A) and (R) are true and (R) is the correct explanation of (A)
208. Match List-I with List-II
- | List-I (Molecule) | List-II (Shape) |
|--------------------------|---------------------------|
| (A) NH_3 | (I) Square pyramid |
| (B) BrF_5 | (II) Tetrahedral |
| (C) PCl_5 | (III) Trigonal pyramidal |
| (D) CH_4 | (IV) Trigonal bipyramidal |
- Choose the correct answer from the options given below : **[JEE-Main-Online 2024]**
 (A) A - III, B - IV, C - I, D - II (B) A - IV, B - III, C - I, D - II
 (C) A - II, B - IV, C - I, D - III (D) A - III, B - I, C - IV, D - II
209. Match List-I and List-II
- | List-I (Hybridization) | List-II (Orientation in Space) |
|-------------------------------|---------------------------------------|
| (A) sp^3 | (I) Trigonal bipyramidal |
| (B) dsp^2 | (II) Octahedral |
| (C) sp^3d | (III) Tetrahedral |
| (D) sp^3d^2 | (IV) Square planar |
- Choose the correct answer from the options given below : **[JEE-Main-Online 2024]**
 (A) A - II, B - I, C - IV, D - III (B) A - IV, B - III, C - I, D - II
 (C) A - III, B - I, C - IV, D - II (D) A - III, B - IV, C - I, D - II

210. Match List-I and List-II

- List-I**
(Molecule / Species)
- (A) SO_2Cl_2
(B) NO
(C) NO_2^-
(D) I_3^-

- List-II**
(Property / Shape)
- (I) Paramagnetic
(II) Diamagnetic
(III) Tetrahedral
(IV) Linear

Choose the correct answer from the options given below :

[JEE-Main-Online 2024]

- (A) A - III, B - I, C - II, D - IV
(C) A - IV, B - I, C - III, D - II

- (B) A - III, B - IV, C - II, D - I
(D) A - II, B - III, C - I, D - IV

211. Match List-I and List-II

- List-I**
(Compound/Species)
- (A) SF_4
(B) BrF_3
(C) BrO_3^-
(D) NH_4^+
- List-II**
(Shape/Geometry)
- (I) Tetrahedral
(II) Pyramidal
(III) See saw
(IV) Bent T-Shape

Choose the correct answer from the options given below :

[JEE-Main-Online 2024]

- (A) A - III, B - IV, C - II, D - I
(C) A - III, B - II, C - IV, D - I

- (B) A - II, B - III, C - I, D - IV
(D) A - II, B - IV, C - III, D - I

212. Match List-I with List-II :

- List-I**
- (A) ICl
(B) ICl_3
(C) ClF_5
(D) IF_7
- List-II**
- (I) T-shape
(II) Square pyramidal
(III) Pentagonal bipyramidal
(IV) Linear

Choose the correct answer from the options given below :

[JEE-Main-Online 2024]

- (A) (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
(C) (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

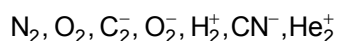
- (B) (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
(D) (A) - (I), (B) - (IV), (C) - (III), (D) - (II)

213. Number of molecules / species from the following having one unpaired electron is _____.



[JEE-Main-Online 2024]

214. The total number of species from the following in which one unpaired electron is present, is _____.

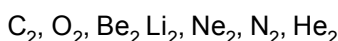


[JEE-Main-Online 2024]

215. Total number of electrons present in (π) molecular orbitals of O_2 , O_2^+ and O_2^- is _____.

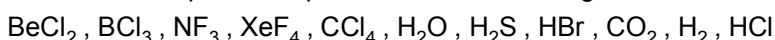
[JEE-Main-Online 2024]

216. Number of molecules having bond order 2 from the following molecules is _____.



[JEE-Main-Online 2024]

217. Number of compounds / species from the following with non-zero dipole moment is _____.

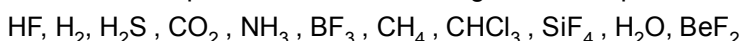


[JEE-Main-Online 2024]

218. In the lewis dot structure for NO_2^- , total number of valence electrons around nitrogen is _____.

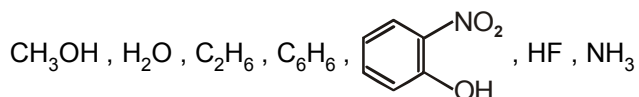
[JEE-Main-Online 2024]

219. Number of compounds from the following with zero dipole moment is _____.



[JEE-Main-Online 2024]

220. Number of molecules from the following which can exhibit hydrogen bonding is _____. (nearest integer)



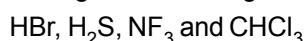
[JEE-Main-Online 2024]

221. Total number of species from the following with central atom utilising sp^2 hybrid orbitals for bonding is ____.
- NH_3 , SO_2 , SiO_2 , BeCl_2 , C_2H_2 , C_2H_4 , BCl_3 , HCHO , C_6H_6 , BF_3 , $\text{C}_2\text{H}_4\text{Cl}_2$ [JEE-Main-Online 2024]

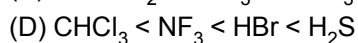
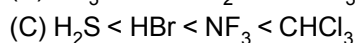
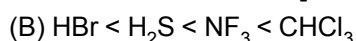
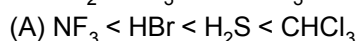
222. Number of molecules from the following which are exceptions to octet rule is ____.
- CO_2 , NO_2 , H_2SO_4 , BF_3 , CH_4 , SiF_4 , ClO_2 , PCl_5 , BeF_2 , C_2H_6 , CHCl_3 , CBr_4 .

[JEE-Main-Online 2024]

223. Arrange the following compounds in increasing order of their dipole moment :



[JEE-Main-Online 2025]



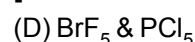
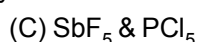
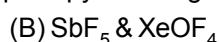
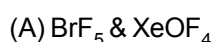
224. An element 'E' has the ionisation enthalpy value of 374 kJ mol^{-1} . 'E' reacts with elements A, B, C and D with electron gain enthalpy values of -328 , -349 , -325 and -295 kJ mol^{-1} , respectively. The correct order of the products EA, EB, EC and ED in terms of ionic character is :

[JEE-Main-Online 2025]



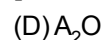
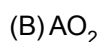
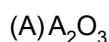
225. The molecules having square pyramidal geometry are

[JEE-Main-Online 2025]



226. The type of oxide formed by the element among Li, Na, Be, Mg, B and Al that has the least atomic radius is :

[JEE-Main-Online 2025]



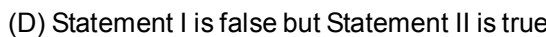
227. Given below are two statements :

[JEE-Main-Online 2025]

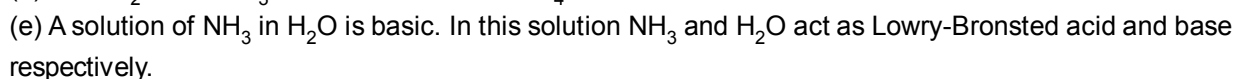
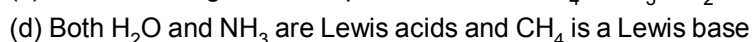
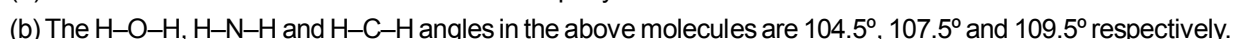
Statement (I) : Experimentally determined oxygen-oxygen bond lengths in the O_3 are found to be same and the bond length is greater than that of a $\text{O}=\text{O}$ (double bond) but less than that of a single ($\text{O}-\text{O}$) bond.

Statement (II) : The strong lone pair-lone pair repulsion between oxygen atoms is solely responsible for the fact that the bond length in ozone is smaller than that of a double bond ($\text{O}=\text{O}$) but more than that of a single bond ($\text{O}-\text{O}$).

In the light of the above statements, choose the **correct** answer from the options given below :

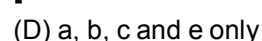
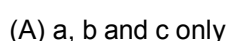


228. Which of the following statement is true with respect to H_2O , NH_3 and CH_4 ?



Choose the correct answer from the options given below :

[JEE-Main-Online 2025]



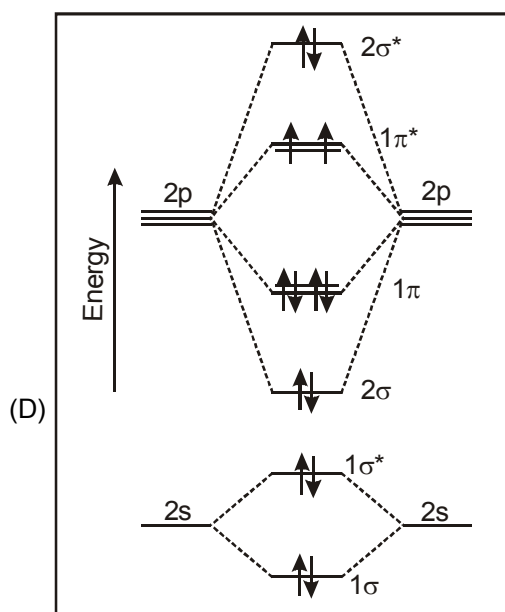
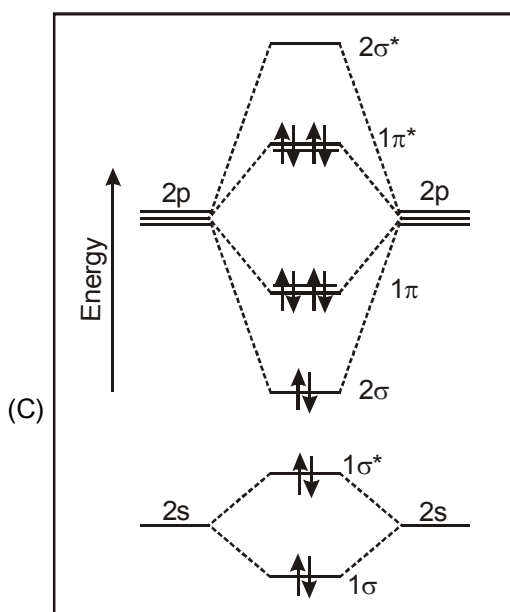
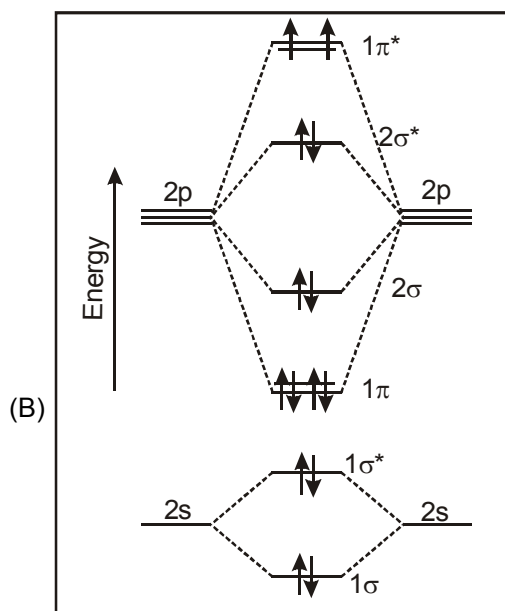
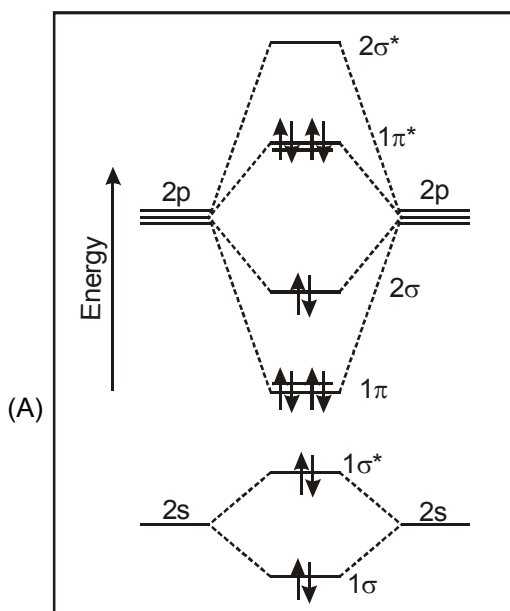
229. Which of the following linear combination of atomic orbitals will lead to formation of molecular orbitals in homonuclear diatomic molecules [internuclear axis in z-direction]? **[JEE-Main-Online 2025]**
- (a) $2p_z$ and $2p_x$ (b) $2s$ and $2p_x$ (c) $3d_{xy}$ and $3d_{x^2-y^2}$ (d) $2s$ and $2p_z$
- (e) $2p_z$ and $3d_{x^2-y^2}$
- (A) e Only (B) a and b Only (C) d Only (D) c and d Only
230. The maximum covalency of a non-metallic group 15 element 'E' with weakest E–E bond is : **[JEE-Main-Online 2025]**
- (A) 5 (B) 3 (C) 6 (D) 4
231. Match the LIST-I with LIST-II
- | LIST-I
(Classification of molecules based on octet rule) | LIST-II
(Example) |
|---|--|
| a. Molecules obeying octet rule | I. NO, NO ₂ |
| b. Molecules with incomplete octet | II. BCl ₃ , AlCl ₃ |
| c. Molecules with incomplete octet with odd electron | III. H ₂ SO ₄ , PCl ₅ |
| d. Molecules with expanded octet | IV. CCl ₄ , CO ₂ |
- Choose the correct answer from the options given below : **[JEE-Main-Online 2025]**
- (A) a-IV, b-II, c-I, d-III (B) a-III, b-II, c-I, d-IV
- (C) a-IV, b-I, c-III, d-II (D) a-II, b-IV, c-III, d-I
232. In Carius method for estimation of halogens, 180 mg of an organic compound produced 143.5 mg of AgCl. The percentage composition of chlorine in the compound is _____. **[JEE-Main-Online 2025]**
- [Given : molar mass in g mol⁻¹ of Ag : 108, Cl = 35.5]
233. Total number of non bonded electrons present in NO₂⁻ ion based on Lewis theory is _____. **[JEE-Main-Online 2025]**

EXERCISE-IV

PREVIOUS YEARS IIT-JEE QUESTIONS

Choose the correct alternative (only one correct answer).

1. The increasing order of atomic radii of the following Group 13 elements is [JEE Advanced 2016]
 (A) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$ (B) $\text{Ga} < \text{Al} < \text{In} < \text{Tl}$
 (C) $\text{Al} < \text{In} < \text{Ga} < \text{Tl}$ (D) $\text{Al} < \text{Ga} < \text{Tl} < \text{In}$
2. The order of the oxidation state of the phosphorus atom in H_3PO_2 , H_3PO_4 , H_3PO_3 and $\text{H}_4\text{P}_2\text{O}_6$ is : [JEE Advanced 2017]
 (A) $\text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2$ (B) $\text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_4 > \text{H}_4\text{P}_2\text{O}_6$
 (C) $\text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6 > \text{H}_3\text{PO}_4$ (D) $\text{H}_3\text{PO}_4 > \text{H}_3\text{PO}_2 > \text{H}_3\text{PO}_3 > \text{H}_4\text{P}_2\text{O}_6$
3. The correct molecular orbital diagram for F_2 molecule in the ground state is [JEE Advanced 2023]



4. Based on **VSEPR** model, match the xenon compounds given in **List-I** with the corresponding geometries and the number of lone pairs on xenon given in **List-II** and choose the correct option.

List-I

- (P) XeF_2
(Q) XeF_4
(R) XeO_3
(S) XeO_3F_2

List-II

- (1) Trigonal bipyramidal and two lone pair of electrons
(2) Tetrahedral and one lone pair of electrons
(3) Octahedral and two lone pair of electrons
(4) Trigonal bipyramidal and no lone pair of electrons
(5) Trigonal bipyramidal and three lone pair of electrons

[JEE Advanced 2024]

- (A) $P \rightarrow 5, Q \rightarrow 2, R \rightarrow 3, S \rightarrow 1$
(C) $P \rightarrow 4, Q \rightarrow 3, R \rightarrow 2, S \rightarrow 1$

- (B) $P \rightarrow 5, Q \rightarrow 3, R \rightarrow 2, S \rightarrow 4$
(D) $P \rightarrow 4, Q \rightarrow 2, R \rightarrow 5, S \rightarrow 3$

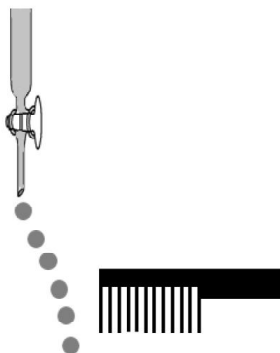
One or more than one correct :

5. The crystalline form of borax has [JEE Advanced 2016]
(A) Tetranuclear $[\text{B}_4\text{O}_5(\text{OH})_4]^{2-}$ unit
(B) All boron atoms in the same plane.
(C) Equal number of sp^2 and sp^3 hybridized boron atoms.
(D) One terminal hydroxide per boron atom.
6. The compound(s) with TWO lone pairs of electrons on the central atom is(are) [JEE Advanced 2016]
(A) BrF_6 (B) ClF_3 (C) XeF_4 (D) SF_4
7. According to Molecular Orbital Theory, [JEE Advanced 2016]
(A) C_2^{2-} is expected to be diamagnetic
(B) O_2^{2+} is expected to have a longer bond length than O_2
(C) N_2^+ and N_2^- have the same bond order
(D) He_2^+ has the same energy as two isolated He atoms
8. The colour of the X_2 molecules of group 17 elements changes gradually from yellow to violet down the group. This is due to [JEE Advanced 2017]
(A) decrease in HOMO-LUMO gap down the group
(B) decrease in $\pi^* - \sigma^*$ gap down the group
(C) the physical state of X_2 at room temperature changes from gas to solid down the group
(D) decrease in ionization energy down the group
9. The correct statement(s) about the oxoacids, HClO_4 and HClO , is(are) [JEE Advanced 2017]
(A) The conjugate base of HClO_4 is weaker base than H_2O
(B) The central atom in both HClO_4 and HClO is sp^3 hybridized
(C) HClO_4 is formed in the reaction between Cl_2 and H_2O
(D) HClO_4 is more acidic than HClO because of the resonance stabilization of its anion
10. The option(s) with only amphoteric oxides is(are) [JEE Advanced 2017]
(A) $\text{Cr}_2\text{O}_3, \text{CrO}, \text{SnO}, \text{PbO}$ (B) $\text{NO}, \text{B}_2\text{O}_3, \text{PbO}, \text{SnO}_2$
(C) $\text{Cr}_2\text{O}_3, \text{BeO}, \text{SnO}, \text{SnO}_2$ (D) $\text{ZnO}, \text{Al}_2\text{O}_3, \text{PbO}, \text{PbO}_2$
11. Among the following, the correct statement(s) is(are) [JEE Advanced 2017]
(A) $\text{Al}(\text{CH}_3)_3$ has the three-centre two-electron bonds in its dimeric structure
(B) AlCl_3 has the three-centre two-electron bonds in its dimeric structure
(C) BH_3 has the three-centre two-electron bonds in its dimeric structure
(D) The Lewis acidity of BCl_3 is greater than that of AlCl_3

12. Based on the compounds of group 15 elements, the correct statement(s) is (are) **[JEE Advanced 2018]**
 (A) Bi_2O_5 is more basic than N_2O_5
 (B) NF_3 is more covalent than BiF_3
 (C) PH_3 boils at lower temperature than NH_3
 (D) The N–N single bond is stronger than the P–P single bond
13. The compound(s) which generate(s) N_2 gas upon thermal decomposition below 300°C is(are) **[JEE Advanced 2018]**
 (A) NH_4NO_3 (B) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ (C) $\text{Ba}(\text{N}_3)_2$ (D) Mg_3N_2
14. Each of the following options contains a set of four molecules. Identify the option(s) where all four molecules possess permanent dipole moment at room temperature. **[JEE Advanced 2019]**
 (A) SO_2 , $\text{C}_6\text{H}_5\text{Cl}$, H_2Se , BrF_5 (B) BeCl_2 , CO_2 , BCl_3 , CHCl_3
 (C) NO_2 , NH_3 , POCl_3 , CH_3Cl (D) BF_3 , O_3 , SF_6 , XeF_6
15. For diatomic molecules, the correct statement(s) about the molecular orbitals formed by the overlap of two $2p_z$ orbitals is(are) **[JEE Advanced 2022]**
 (A) σ orbital has a total of two nodal planes.
 (B) σ^* orbital has one node in the xz-plane containing the molecular axis.
 (C) π orbital has one node in the plane which is perpendicular to the molecular axis and goes through the center of the molecule.
 (D) π^* orbital has one node in the xy-plane containing the molecular axis.
16. The compound(s) having peroxide linkage is(are) **[JEE Advanced 2024]**
 (A) $\text{H}_2\text{S}_2\text{O}_7$ (B) $\text{H}_2\text{S}_2\text{O}_8$ (C) $\text{H}_2\text{S}_2\text{O}_5$ (D) H_2SO_5
17. The option(s) in which at least three molecules follow Octet Rule is(are) **[JEE Advanced 2024]**
 (A) CO_2 , C_2H_4 , NO and HCl (B) NO_2 , O_3 , HCl and H_2SO_4
 (C) BCl_3 , NO , NO_2 and H_2SO_4 (D) CO_2 , BCl_3 , O_3 and C_2H_4
18. The correct statement(s) about intermolecular forces is(are) **[JEE Advanced 2025]**
 (A) The potential energy between two point charges approaches zero more rapidly than the potential energy between a point dipole and a point charge as the distance between them approaches infinity.
 (B) The average potential energy of two rotating polar molecules that are separated by a distance r has $1/r^3$ dependence.
 (C) The dipole-induced dipole average interaction energy is independent of temperature.
 (D) Nonpolar molecules attract one another even though neither has a permanent dipole moment.
19. Regarding the molecular orbital (MO) energy levels for homonuclear diatomic molecules, the **INCORRECT** statement(s) is(are) **[JEE Advanced 2025]**
 (A) Bond order of Ne_2 is zero.
 (B) The highest occupied molecular orbital (HOMO) of F_2 is σ -type.
 (C) Bond energy of O_2^+ is smaller than the bond energy of O_2 .
 (D) Bond length of Li_2 is larger than the bond length of B_2 .

Integers Type :

20. The sum of the number of lone pairs of electrons on each central atom in the following species is :
[TeBr₆]²⁻, [BrF₂]⁺, SNF₃, and [XeF₃]⁻
(Atomic numbers : N = 7, F = 9, S = 16, Br = 35, Te = 52, Xe = 54) [JEE Advanced 2017]
21. Among H₂, He₂⁺, Li₂, Be₂, B₂, C₂, N₂, O₂⁻, and F₂, the number of diamagnetic species is
(Atomic numbers : H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8, F = 9) [JEE Advanced 2017]
22. The total number of compounds having at least one bridging oxo group among the molecules given below is _____.
N₂O₃, N₂O₅, P₄O₆, P₄O₇, H₄P₂O₅, H₅P₃O₁₀, H₂S₂O₃, H₂S₂O₅ [JEE Advanced 2018]
23. Among B₂H₆, B₃N₃H₆, N₂O, N₂O₄, H₂S₂O₃ and H₂S₂O₈, the total number of molecules containing covalent bond between two atoms of the same kind is _____. [JEE Advanced 2019]
24. Consider the following compounds in the liquid form :
O₂, HF, H₂O, NH₃, H₂O₂, CCl₄, CHCl₃, C₆H₆, C₆H₅Cl.
When a charged comb is brought near their flowing stream, how many of them show deflection as per the following figure ? [JEE Advanced 2020]



25. Among [I₃]⁺, [SiO₄]⁴⁻, SO₂Cl₂, XeF₂, SF₄, ClF₃, Ni(CO)₄, XeO₂F₂, [PtCl₄]²⁻, XeF₄, and SOCl₂, the total number of species having sp³ hybridised central atom is _____. [JEE Advanced 2023]
26. Consider the following molecules: Br₃O₈, F₂O, H₂S₄O₆, H₂S₅O₆, and C₃O₂.
Count the number of atoms existing in their zero oxidation state in each molecule.
Their sum is _____. [JEE Advanced 2023]

ANSWERS KEY

EXERCISE-I

1.	A	2.	B	3.	C	4.	C	5.	A	6.	B	7.	C
8.	A	9.	A	10.	D	11.	C	12.	D	13.	D	14.	C
15.	C	16.	D	17.	C	18.	A	19.	B	20.	C	21.	A
22.	D	23.	C	24.	D	25.	B	26.	B	27.	D	28.	B
29.	D	30.	A	31.	C	32.	D	33.	C	34.	B	35.	C
36.	C	37.	C	38.	A	39.	B	40.	D	41.	A	42.	A
43.	D	44.	B	45.	B	46.	B	47.	A	48.	C	49.	B
50.	B	51.	B	52.	C	53.	C	54.	B	55.	B	56.	C
57.	D	58.	C	59.	B	60.	D	61.	D	62.	C	63.	C
64.	C	65.	C	66.	B	67.	D	68.	A	69.	C	70.	C
71.	D	72.	B	73.	B	74.	C	75.	B	76.	B	77.	B
78.	A	79.	C	80.	A	81.	C	82.	C	83.	B	84.	A
85.	B	86.	D	87.	C	88.	B	89.	A	90.	C	91.	C
92.	D	93.	B	94.	A	95.	A	96.	C	97.	B	98.	A
99.	D	100.	A	101.	C	102.	B	103.	D	104.	A	105.	B
106.	B	107.	C	108.	D	109.	B	110.	A	111.	C	112.	A
113.	C	114.	A	115.	C	116.	C	117.	5.00	118.	4.00	119.	4.00
120.	0.00	121.	4.00	122.	3.00	123.	4.00	124.	3.00	125.	13.00	126.	4.00

EXERCISE-II

01. B	02. C	03. B	04. B	05. B
06. D	07. A	08. C	09. C	10. C
11. B	12. D	13. B	14. A	15. B
16. D	17. B	18. A	19. D	20. B
21. A	22. B	23. A	24. A	25. D
26. A,B	27. A,B,D	28. A,B	29. C,D	30. A,C,D
31. A,D	32. C,D	33. A,B	34. A	35. D
36. B	37. C			
38. A $\rightarrow$ R ; B $\rightarrow$ P ; C $\rightarrow$ T ; D $\rightarrow$ S		39. A $\rightarrow$ T ; B $\rightarrow$ P ; C $\rightarrow$ Q ; D $\rightarrow$ R		
40. A $\rightarrow$ R ; B $\rightarrow$ S ; C $\rightarrow$ P ; D $\rightarrow$ Q		41. A $\rightarrow$ S ; B $\rightarrow$ T ; C $\rightarrow$ Q ; D $\rightarrow$ P		
42. A $\rightarrow$ R ; B $\rightarrow$ P ; C $\rightarrow$ Q				

EXERCISE-III

1.	A	2.	B	3.	C	4.	D	5.	A	6.	A	7.	D
8.	B	9.	D	10.	D	11.	C	12.	D	13.	A	14.	B
15.	D	16.	D	17.	A	18.	D	19.	19.00	20.	3.00	21.	A
22.	C	23.	D	24.	C	25.	B	26.	C	27.	D	28.	C
29.	B	30.	B	31.	D	32.	D	33.	D	34.	C	35.	B

36.	A	37.	D	38.	C	39.	D	40.	B	41.	B	42.	C
43.	C	44.	D	45.	A	46.	B	47.	B	48.	B	49.	D
50.	B	51.	C	52.	C	53.	A	54.	C	55.	B	56.	D
57.	173.00	58.	2.00	59.	03.00	60.	00.00	61.	0.00	62.	10.00	63.	3.00
64.	10.00	65.	2.00	66.	C	67.	B	68.	B	69.	A	70.	A
71.	C	72.	C	73.	B	74.	C	75.	C	76.	B	77.	C
78.	A	79.	B	80.	B	81.	A	82.	C	83.	A	84.	C
85.	C	86.	A	87.	A	88.	D	89.	B	90.	B	91.	A
92.	C	93.	C	94.	D	95.	A	96.	C	97.	A	98.	B
99.	A	100.	C	101.	B	102.	D	103.	B	104.	B	105.	B
106.	3.00	107.	3.00	108.	2.00	109.	4.00	110.	1.00	111.	4.00	112.	4.00
113.	3.00	114.	6.00	115.	3.00	116.	1.00	117.	0.00	118.	2.00	119.	C
120.	C	121.	C	122.	B	123.	A	124.	C	125.	C	126.	A
127.	D	128.	C	129.	A	130.	B	131.	B	132.	C	133.	D
134.	D	135.	C	136.	C	137.	C	138.	A	139.	C	140.	C
141.	B	142.	D	143.	A	144.	B	145.	C	146.	D	147.	B
148.	A	149.	C	150.	C	151.	06.00	152.	5.00	153.	03.00	154.	17.00
155.	03.00	156.	3.00	157.	04.00	158.	01.00	159.	03.00	160.	04.00	161.	05.00
162.	04.00	163.	12.00	164.	8.00	165.	3.00	166.	6.00	167.	3.00	168.	5.00
169.	5.00	170.	4.00	171.	B	172.	A	173.	B	174.	B	175.	D
176.	C	177.	B	178.	D	179.	A	180.	A	181.	D	182.	C
183.	A	184.	3.00	185.	4.00	186.	4.00	187.	8.00	188.	1.00	189.	4.00
190.	6.00	191.	3.00	192.	4.00	193.	6.00	194.	0.00	195.	B	196.	C
197.	C	198.	C	199.	A	200.	B	201.	C	202.	C	203.	A
204.	A	205.	D	206.	D	207.	D	208.	D	209.	D	210.	A
211.	A	212.	B	213.	2.00	214.	4.00	215.	6.00	216.	2.00	217.	5.00
218.	8.00	219.	6.00	220.	5.00	221.	6.00	222.	6.00	223.	A	224.	C
225.	A	226.	A	227.	A	228.	A	229.	C	230.	D	231.	A
232.	20.00	233.	12.00										

EXERCISE-IV

Choose the correct alternative (only one correct answer) :

1. B 2. A 3. C 4. B

One or more than one correct :

5. A,C,D 6. B,C 7. A,C 8. AB 9. ABD 10. C,D 11. A,C,D
12. A,B,C 13. B,C 14. A, C 15. A, D 16. B,D 17. A,D 18. B, D
19. B, C

Integers Type :

20. 6 21. 6 22. 5 or 6 23. 4 24. 6 25. 5.00 26. 6.00